

Chapter 45

Working with Audio in the Edit Page

DaVinci Resolve has a solid set of features for editing, mixing, and mastering audio in your programs right in the Edit page.

Whether you're adjusting synced audio for dailies, finessing the levels of an edited project you're assembling, mixing a program for output, or importing and laying in audio mix files from the sound designer to output for mastering, DaVinci Resolve has Level, Pan, and Channel Assignment controls to control your audio output for both monitoring and delivery, automated fader recording at both the track and clip level for mixing, and VST and Audio Unit audio filter support for mastering audio tracks and channels using industry-standard noise reduction, compression, EQ, and other filters. And, if all that's not enough for you, you can export to Pro Tools in the Delivery page to hand off your program and its audio in a state ready for further work.

When you've finished doing the editorial audio work in your program, and you want to really drill into your program's audio for detailed audio editing and mixing, you can use the audio-specific tools of the Fairlight page.

For more information, see *Chapter 170, "Using the Fairlight Page."*

Contents

Audio in the Edit vs. Fairlight Pages	951
Compatible Audio Formats	951
Assigning Audio Channels in the Media Pool	951
How to Assign Audio Channels	952
Support for Mixed Audio Track Formats from Source Clips	955
Mono-Multichannel Audio Files with Suffixes Import as a Multichannel Clip	956
Switch Mono Audio Channels of a Clip on the Timeline	956
Editing Audio Into the Timeline	957
Editing Audio Using the Source Viewer.....	957
Simultaneous Audio Waveform Display in the Source Viewer.....	958
Using Multi-Channel Timeline Tracks	959
Creating Timelines With Audio Mixer Presets	960

Editing Audio Clips Into the Timeline.....	961
Changing Audio Clip Attributes After Editing.....	962
Displaying Waveforms in the Timeline.....	962
Editing Audio In the Timeline Using In and Out Points.....	963
Resizing Audio Clips in Subframe Increments.....	965
Enabling and Disabling Subframe Editing.....	965
Subframe Editing of In and Out Points.....	965
Eliminating Subframe Audio Edits.....	966
Record Voiceover Tool.....	966
Audio Settings in the Inspector.....	967
Setting Volume.....	967
Adjusting Audio in the Inspector.....	967
Adjusting Audio in the Timeline.....	968
Adjusting Volume Using Keyboard Shortcuts.....	969
Normalize Audio Volume Command.....	969
Keyframing Audio.....	970
Overlay Controls for Volume.....	971
How to Add and Adjust Volume Keyframes.....	971
Audio Fade Handles.....	972
Audio Crossfades.....	973
Voice Isolation (Studio Version Only).....	974
Dialogue Leveler.....	975
Music Remixer (Studio Version Only).....	977
Music Editor (Studio Version Only).....	977
Dialogue Separator Track FX (Studio Version Only).....	979
Ducker Track FX.....	979
Beat Detector (Studio Version Only).....	982
Dialogue Matcher (Studio Version Only).....	982
Audio Assistant (Studio Version Only).....	983
The Audio Mixer.....	984
Audio Mixer Controls.....	985
Mute and Solo Tracks For Output.....	985
Displaying Audio Meters.....	985
Audio Compound Clips.....	986
Audio Playback for Variable Speed Clips.....	986
Using Audio Filters.....	986
Installing Audio Filters.....	988
Voice Convert (Studio Version Only).....	988
Using Voice Convert.....	989
Voice Convert Settings.....	990
Creating Your Own Voice Models.....	990
The Fairlight Page.....	992

Audio in the Edit vs. Fairlight Pages

While the Fairlight page provides dedicated audio editing and mixing capabilities that are suitable for sweetening the audio of your program once it's been edited, the Edit page has extensive audio capabilities of its own. This enables editors to edit and refine audio clips, set levels, and do simple mixes as they assemble the program in the first place. However, once things have been edited together, you're meant to go freely back and forth between the Edit and Fairlight page as you refine your work, using whichever environment is most suitable for the task at hand.

Compatible Audio Formats

DaVinci Resolve is compatible with WAVE, Broadcast WAVE, AIFF, MP3, AAC (M4A), CAF, iOS Voice Memo (macOS only), both MTS and QuickTime containers that use the AC3 audio format, and Enhanced AC-3 (macOS and Windows only). DaVinci Resolve is compatible with audio at sample rates including 32, 44.1, 48, 88.2, 96, and 192 kHz.

Assigning Audio Channels in the Media Pool

When you first import audio into the Media Pool, it's a good idea to make sure that whatever channels those files contain are assigned correctly before you start editing clips into the Timeline. In other words, you want to make sure that stereo files are set to be stereo clips, that 5.1 and 7.1 files are set to be surround clips, and that multi-channel files are set to expose however many tracks you want to edit separately in your program. Clip channel assignments are made in the Audio panel of the Clip Attributes window.

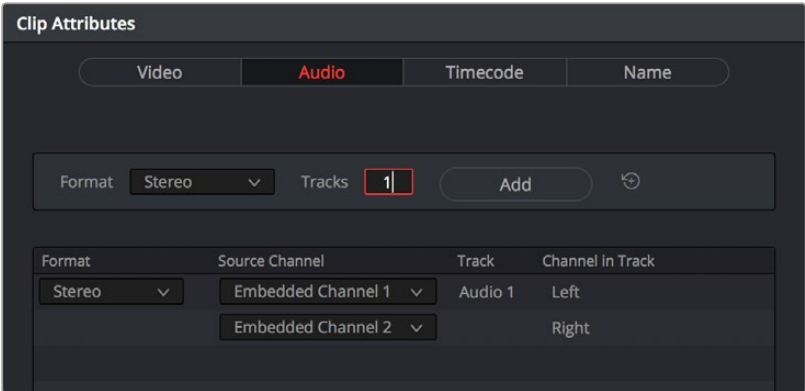
This is particularly important when clips have more than two channels of audio. For example, production sound recordists might record three, six, or even more audio channels, corresponding to multiple microphones used on set to simultaneously record different actors plus a mixdown track. In this case, you need to define how many of these channels you want to play (or mute), and how many audio items you want to appear in the Edit and Fairlight page Timelines.

Mono, stereo, 5.1, and 7.1 clips are handled automatically, but multi-channel clips needing custom assignments in the Media Pool should be remapped as necessary using Clip Attributes, so that DaVinci Resolve can more easily place incoming audio clips into the correct track of the Timeline. You can alter the clip attributes for clips one at a time, or for multiple selected clips at once.

For more information on these settings, see *Chapter 22, "Modifying Clips and Clip Attributes."*

How to Assign Audio Channels

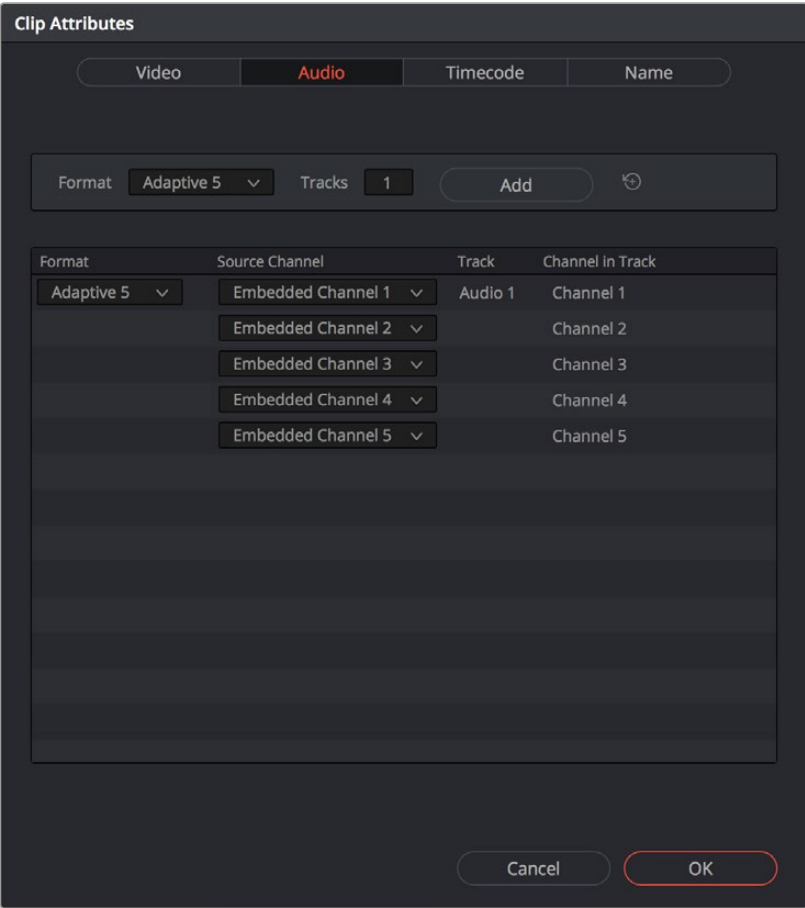
Each clip with audio has the following options in the Audio panel of the Clip Attributes window:



The controls available for adding tracks with which to remap channels

A set of controls at the top of the Track/Channel list lets you add additional tracks to a clip. Adding additional tracks to a clip let you remap that clip’s available channels to appear as additional items in the timeline when you edit it, one item per track with an unmuted channel.

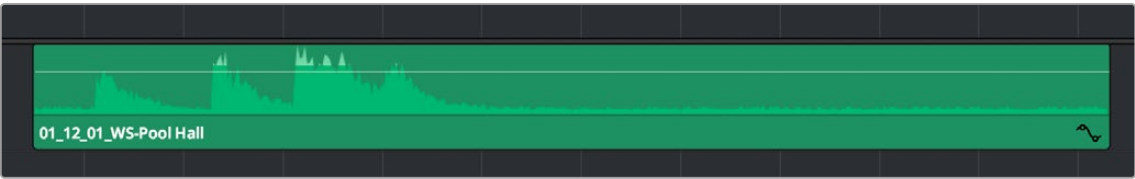
How you handle audio in your timeline is up to you. In the case of production audio consisting of five channels, four for different microphones plus one mixdown channel, you might use Clip Attributes to map all channels to a single track.



The Audio panel of the Clip Attributes window for a single-track clip with four channels

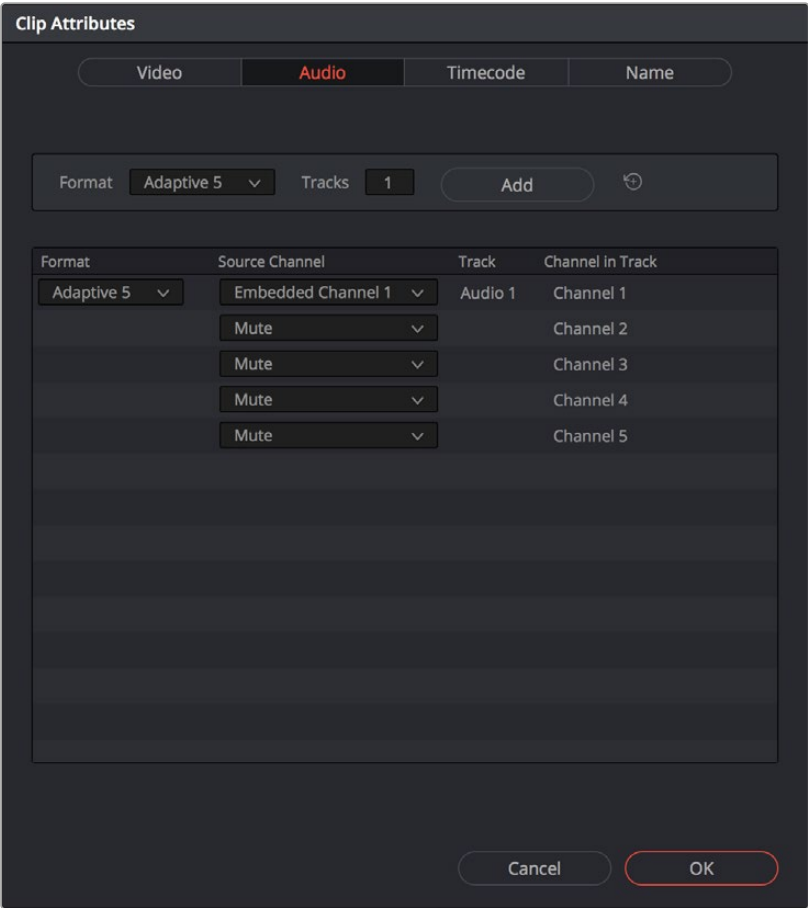
EDIT

With this mapping, this audio clip exposes only one item on one track in the Edit page Timeline, or five lanes within a single track in the Fairlight page Timeline. Either way, this mapping exposes a single editable unit.



A single audio channel exposed in the Timeline

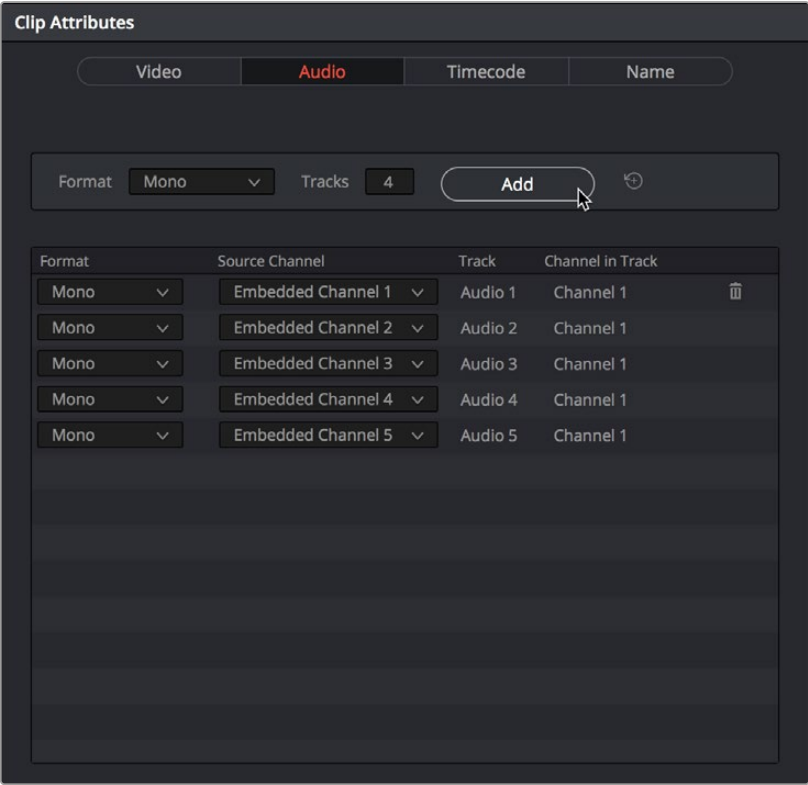
Optionally, you could choose to mute the four individual microphone channels and only monitor the top mixdown channel. This can be accomplished by setting channels 2-5 to mute. Muted channels are always retained, just not heard, and you can turn them back on whenever you like using by right-clicking a clip in the Timeline and choosing Clip Attributes to get back to these controls for the specific timeline clip you’ve edited.



The Audio panel of the Clip Attributes window for a clip with four channels muted

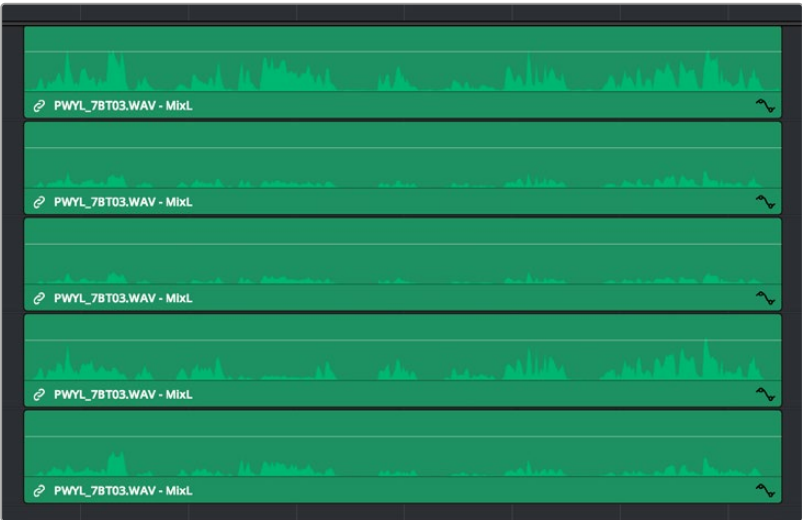
NOTE: If you export audio clips with muted tracks to Pro Tools in the Deliver page, the muted channels are exported as well.

Alternately, you might elect to set these clips up with five separate audio tracks with a single audio channel each, in order to expose each channel separately in the Timeline for independent editing. First, set the top track to be Mono using Embedded Channel 1. Then, using the Add channel controls at the top of the channels list, you can set Tracks to 4, format to Mono, and click Add to create four additional tracks in addition to the one track that clip had originally. Remapping channels 2-5 to these new tracks will result in the track mapping seen in the following screenshot.



The Audio panel of the Clip Attributes window after adding four channels

Editing such a clip into the Timeline results in five linked audio items appearing on five separate audio tracks, each of which can be edited separately in the Timeline.

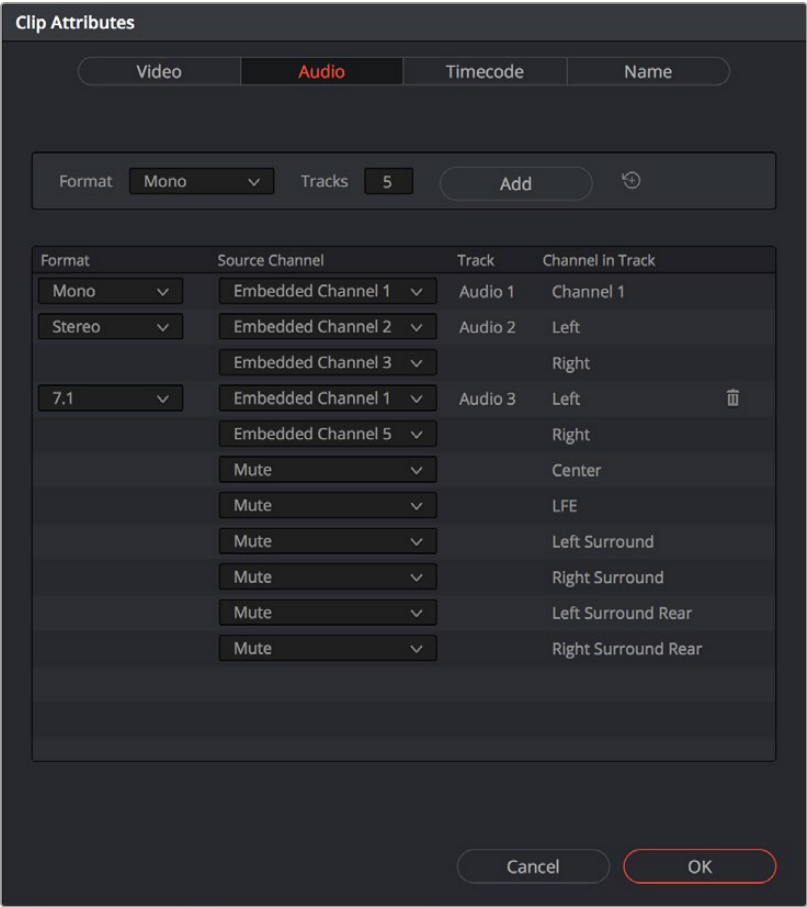


Each channel exposed as a separately editable clip in the Timeline

Support for Mixed Audio Track Formats from Source Clips

DaVinci Resolve also supports media with multiple audio tracks that have differently formatted channels embedded within them. For example, a clip with one stereo track, one 5.1 surround track, and six mono tracks can all be appropriately set up in the Audio panel of Clip Attributes after that clip has been imported.

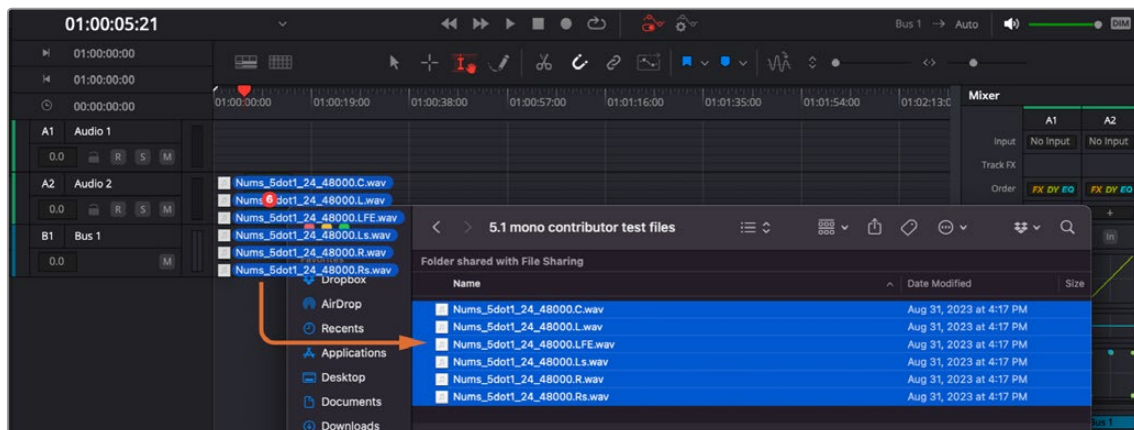
The Audio panel of Clip Attributes now has controls over what format (Mono, Stereo, 5.1, 7.1, Adaptive) the channels embedded within a particular clip should be configured as. This means that you can set up clips with multiple tracks, each one using potentially different formats of audio employing different combinations of clips, which is handy for mastering.



Clip Attributes now lets you assign channels among different tracks with different channel assignments.

Mono-Multichannel Audio Files with Suffixes Import as a Multichannel Clip

DaVinci Resolve can interpret and import mono audio files with channel name suffixes as a single formatted multichannel audio clip. The format depends on the number of files detected and the channel suffixes. As an example, Audio001.L.wav and Audio001.R.wav by themselves are interpreted as a stereo 'Audio001.wav' clip. A third file with a 'C' or 'LFE' suffix results in LCR or 2.1 formatted media.



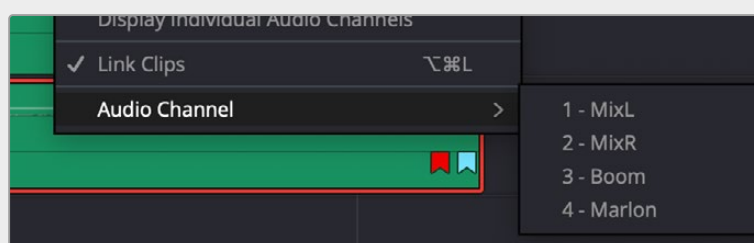
Importing multi-mono audio files to a single multichannel track

Switch Mono Audio Channels of a Clip on the Timeline

You can change your mic (or other) audio sources on mono channel lanes in the Edit page by using a context menu or via the Inspector. This can be handy when editing using a mix track, and you find there is a lav mic that has noise on it, and you want to use boom audio instead.

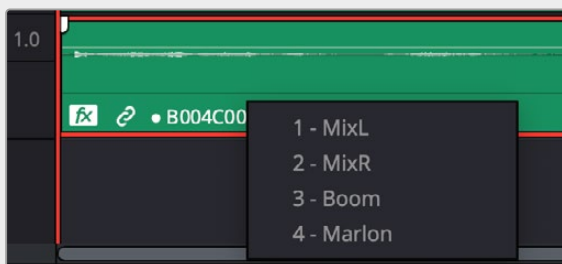
There are two methods for changing mono audio sources: in the timeline or in the Inspector.

Changing mono audio sources via context menu on the timeline:



Right-clicking on an audio clip will reveal the source audio channel choices to choose from

If you are on a mono timeline track and the parent audio file for your track contains multiple mono tracks, right-click on it and an Audio Channel choice appears at the bottom of the context menu. You can then choose any mono source from the hierarchical menu.

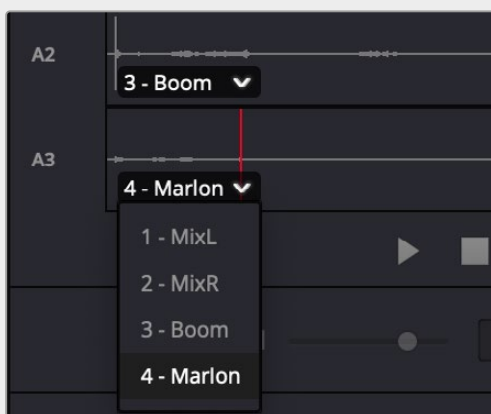


Command right-click on an audio clip in the timeline to reveal just the source audio channels

Alternatively, for an even more streamlined workflow, you can press Command right-click and you will then get a drop-down with only the mono sources:

Changing mono audio sources via the Inspector:

When a timeline clip is opened via the File Inspector, you can change mono track/lane sources by using a drop-down on the channel waveform.



You can change the audio channel by using a drop-down menu on the channel in the File Inspector in the Audio Configuration section.

Editing Audio Into the Timeline

A separate set of audio tracks in the Edit page Timeline contain all of the audio that you edit into the Timeline, as well as any stand-alone audio files that might have been imported along with an AAF or XML file.

Editing Audio Using the Source Viewer

Opening an audio-only clip into the Source Viewer, or opening a clip with both video and audio and setting the Viewer to Audio Waveform results in a split view, with the complete waveform of the entire source clip shown in the top half, and a zoomed-in view of the waveform in the bottom half that can be set to zoom from 1x to 50x from the Zoom menu at the upper left-hand corner of the Source Viewer. This view makes it easy to drag the box at the top to find the section of audio you need relative to the entire clip, and yet still place In and Out points with great precision using the scrubber bar below.

This view shows every channel within each track of the current clip.

You can add markers and set In and Out points for audio clips just as you would for any other clip, in preparation for editing.

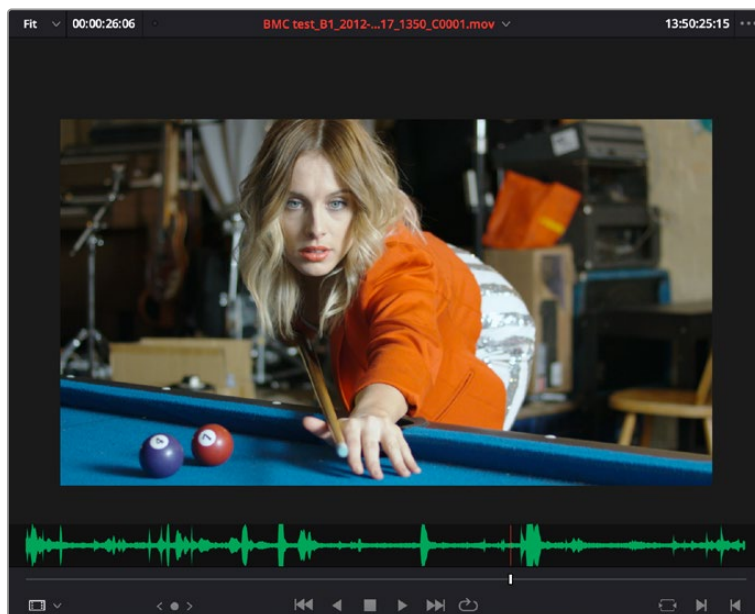


An audio clip opened into the Source Viewer

Simultaneous Audio Waveform Display in the Source Viewer

It's also possible to edit using audio waveforms even when the Source Viewer is set to Source. Two options in the Option Menu let you see a superimposed audio waveform running along the bottom of the Viewer, over the video of the currently selected clip.

- **Show Current Frame Audio Waveform:** Shows a zoomed-in section of audio that scrolls as you play the clip. Useful for seeing dialog and music cues as you play through a clip.
- **Show Full Clip Audio Waveform:** Shows the audio waveform for the entire source media of that clip. The section of audio from the In to Out points you've set in the Source Viewer are highlighted. Useful for using the audio waveform to navigate throughout that clip using the waveform as a reference.



The Source Viewer with "Show Current Frame Audio Waveform" enabled, displaying the audio waveform along the bottom of the image

Using Multi-Channel Timeline Tracks

Multi-channel audio tracks in the Edit page Timeline are extremely convenient when you're dealing with clips that are stereo, 5.1, 7.1, or have an arbitrary number of channels that were recorded in the field, as you can fit all of these channels as a single clip into a single track, that will be correctly mapped to your project's outputs, and that can be edited conveniently as a single item in the Timeline.

However, when you open the Fairlight page you'll see that although the overall number of audio tracks is identical to the Edit page, the Fairlight page shows the channels that are otherwise hidden on the Edit page as lanes within each track, which expose each channel as a visible audio item in the Timeline. In this way, editors can work with multi-channel audio without worrying about visual clutter, while audio editors and mixers can see every channel on every track to help them get their work done.

Getting back to audio tracks on the Edit page, there are different types of audio tracks just like there are different types of audio clips: Mono, Stereo, 5.1, 7.1, and Adaptive. While you can edit any kind of audio clip into any kind of audio track, all clip audio channels that exceed the number of channels possessed by a particular type of timeline track will be muted. For example, you're allowed to edit a six-channel Adaptive audio clip into a Mono audio track, but only the first channel will play because the Mono track only outputs one channel.

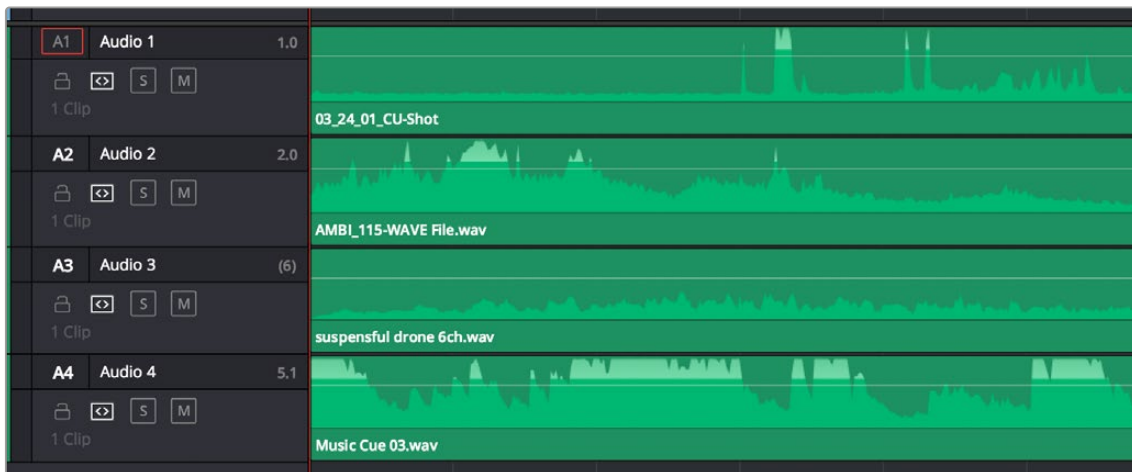
Because of this, it's a good idea to organize your timeline such that all clips appear on tracks that can accommodate the number of channels they have.

TIP: Editing an audio clip into the undefined gray area below existing audio tracks in the Timeline will result in the automatic creation of new audio tracks that are equal in number to the number of tracks the clip you're dragging has, and each new track will have audio mappings that match the incoming audio items.

Defining Timeline Audio Track Channels at Creation

When you first create a new audio track, you have to choose what kind of audio track it will be. Right-clicking in the bottom audio portion of the Timeline track header reveals a contextual sub-menu that lets you create one of three different kinds of audio tracks.

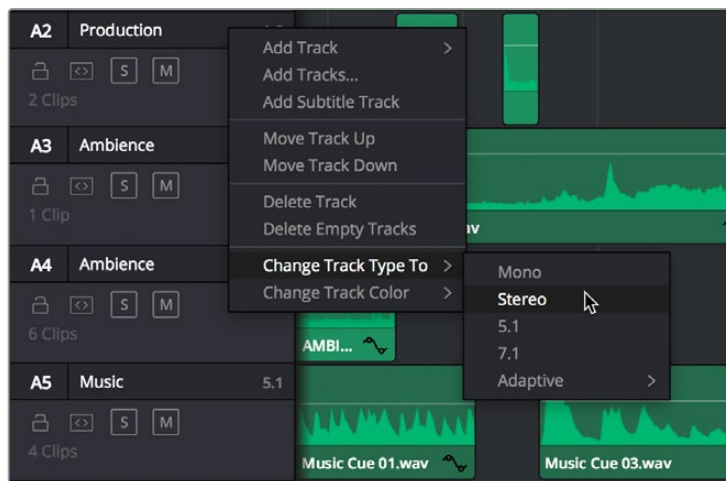
- **Mono:** Holds a single channel.
- **Stereo:** Holds stereo left and right channels. Stereo tracks can be panned.
- **5.1:** Holds the six channels corresponding to a 5.1 surround mix. For broadcast, SMPTE specifies Left, Right, Center, LFE, Left Surround, Right Surround. For cinema distribution these tracks are ordered Left, Center, Right, Surround Left, Surround Right, and LFE.
- **7.1:** Holds the eight channels corresponding to a 7.1 surround mix. For broadcast, SMPTE specifies Left, Right, Center, LFE, Left Surround, Right Surround, Back Surround Left, and Back Surround Right. For cinema distribution these tracks are ordered Left, Center, Right, Left Surround, Right Surround, LFE, Back Left Surround, and Back Right Surround.
- **Adaptive:** Capable of holding up to 24 audio channels. An adaptive audio track can hold clips with different combinations of channels, up to the maximum number of channels allowed within that track. The number of channels allowable on a particular Adaptive track is user-definable (1–24) at the time that track is created. If you edit a clip with more channels into an Adaptive track that was created to hold fewer channels, the extra channels are muted.



Four audio tracks with a variety of audio tracks shown. From the top down, Mono, Stereo, Adaptive, 5.1

Changing How Many Channels an Audio Track Has

If you had set up your timeline with one kind of audio track, but you discover you actually need a different kind, you can change any audio track's type at any time. Just right-click anywhere in that audio track's timeline header, and choose an option from the Change Track Type To submenu of the contextual menu.



Contextual menu for changing audio track types

Creating Timelines With Audio Mixer Presets

For advanced audio workflows and ease of use with Fairlight, you can now create a timeline with pre-assigned audio tracks using a previously created Fairlight Configuration preset. To use this function, create a new timeline, and check the "Use Fairlight Configuration Preset" box. A drop-down menu then appears, allowing you to select the specific preset for the Timeline.

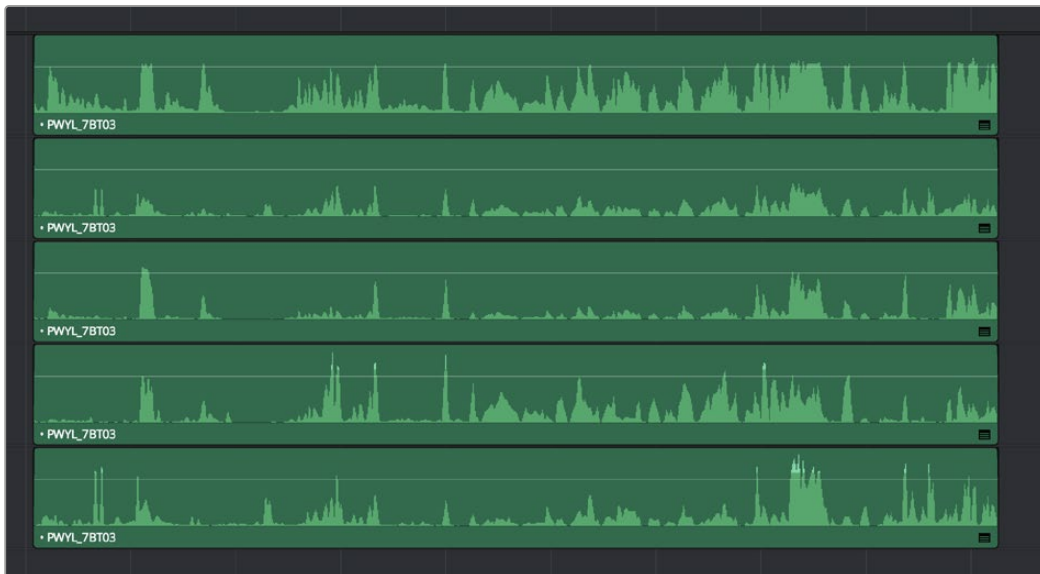
You can create Fairlight Configuration presets using the Fairlight Presets Library, available from the Fairlight menu.

For more information, see *Chapter 171, "Setting Up Tracks, Busses, and Patching."*

Editing Audio Clips Into the Timeline

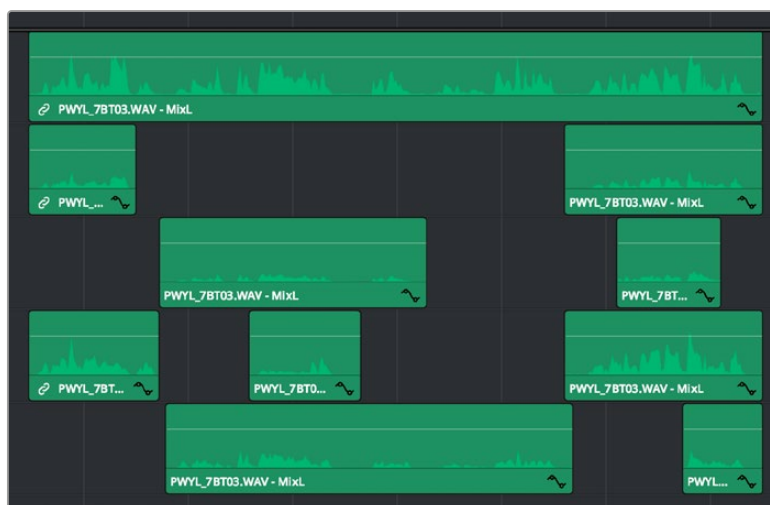
When you edit a video clip with accompanying audio, or an audio-only clip, into the Edit page Timeline, what you see depends on how the audio's internal tracks and channels were defined in the Media Pool, using Clip Attributes. If you've defined a clip to expose multiple tracks of audio, each exposing a different channel, then you exchange the convenience of managing multiple channels of audio as a single item for the freedom to individually edit each channel of audio separately, as individual clips in the Timeline.

For example, if you've been given a multi-channel recording that consists of two boom microphones, two separate lavalier microphones, and a mixdown track that were recorded simultaneously, you can use the Audio panel of the Clip Attributes window to set that clip's audio up as 5-channel Adaptive audio with 5 tracks containing one channel each. Editing this into the Timeline, you end up with five separate audio items appearing in five tracks.



Editing a multi-channel production recording as five separate tracks of audio

This way, when you edit that clip into the Timeline, each audio channel appears as its own clip in its own audio track of the Timeline, which can be separately edited so you can edit the scene to isolate the best dialog from each microphone.



Editing multi-track audio to isolate the best dialog from each microphone

Changing Audio Clip Attributes After Editing

It's best to make decisions about which audio tracks and channels are assigned prior to beginning editing. This is because once you've edited a clip into a timeline, you can't use the Clip Attributes window to edit how many audio tracks and audio channels are exposed in the Timeline.

However, you can use Clip Attributes to change which channels are assigned and/or muted within the available tracks and channels you've edited into the Timeline. For example, if you're editing clips that have five channels of source audio (channels 1 and 2 are a stereo mix and channels 3 through 5 are three different microphones), you may have set your synced source clips to have one audio track and five audio channels, with channels 3–5 muted. Later you have a few clips that would sound much better if you only used channel 4, which is the isolated lavalier microphone for that actor, so you can select those clips and use the Audio panel of Clip Attributes to mute all channels but channel 4.

If, for whatever reason, you need to expose more audio tracks in the Timeline than you originally set an audio clip to use, you can do the following.

To re-edit an audio clip to expose more audio tracks than were originally available:

- 1 Right-click the clip you want to change the audio track mapping of in the Timeline, and choose Find in Media Pool from the contextual menu.
- 2 Right-click that clip in the Media Pool and choose Clip Attributes from the contextual menu.
- 3 Open the Audio panel of the Clip Attributes dialog, and choose how many audio tracks and audio channels you want to set that clip up with. Click OK.
- 4 Once that's done, edit the changed audio clip from the Media Pool to the Timeline to replace the original clip using whichever method makes sense.

Displaying Waveforms in the Timeline

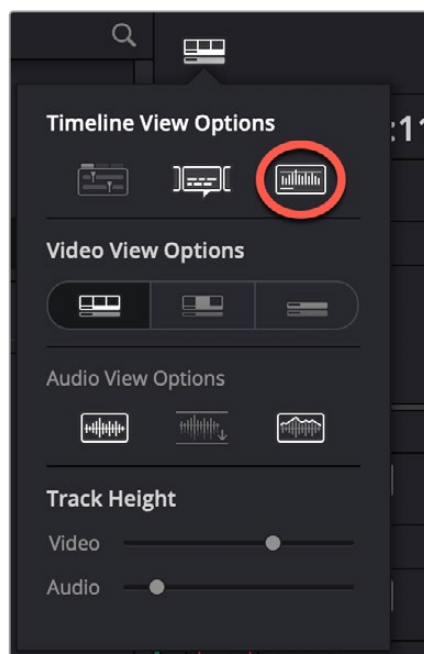
The Timeline View options palette lets you turn Audio Waveform display on and off via a button at the top. The Audio View options let you define how you wish your waveforms to be presented on the Timeline.

Audio View Options: Three buttons govern the look of audio waveforms in the Timeline, when visible.

- **Non-Rectified Waveform:** Lets you toggle between the waveform being drawn from the bottom of the audio track up, or centered and mirrored about itself.
- **Full Waveform:** Hides the divider bar that keeps the waveform separate from the file name area of each audio clip, so the waveform occupies the full space of each audio bar in the Timeline.
- **Waveform Border:** Draws a dark border around the edges of each waveform to make them easier to see.

Video track height slider: Lets you resize the size of all video tracks at once, independently of the audio tracks.

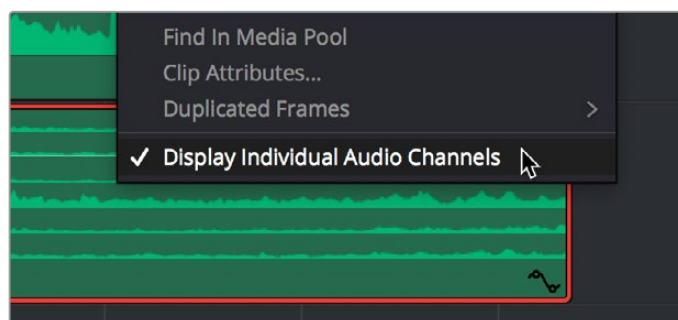
Audio track height slider: Lets you resize the size of all audio tracks at once, independently of the video tracks.



The Audio Waveform display option (circled in red) in the Timeline View drop-down. Audio View Options are (L-R) Non-Rectified Waveforms, Full Waveform, and Waveform Border.

While a single averaged audio waveform representing all the channels in that clip is shown by default, you can switch any clip to seeing each individual waveform in a vertical stack by right-clicking any audio clip and choosing Display Individual Audio Channels.

Whenever you cut an audio clip, you cut all audio channels with it. Audio channels that are embedded within a single track cannot be individually edited.



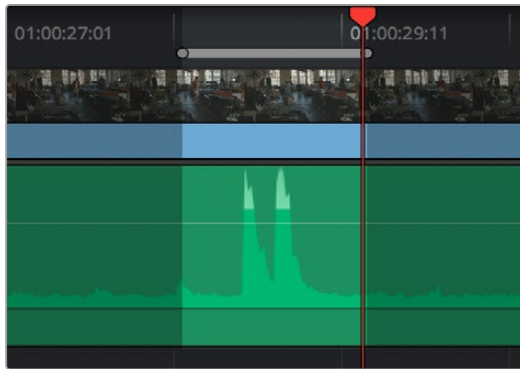
Enabling the display of multiple channel waveforms in the Timeline

Editing Audio In the Timeline Using In and Out Points

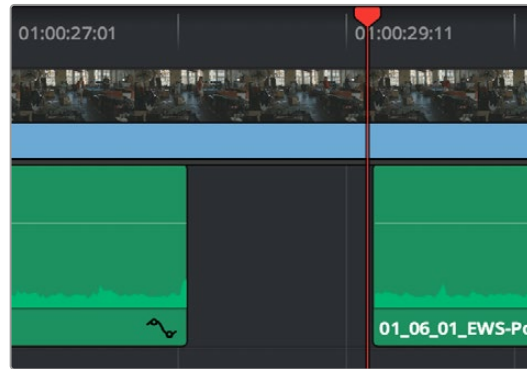
Audio clips can be edited using all of the commands and tools available for video clips. However, it's good to know that one of the most common techniques of editing audio in other environments is available in DaVinci Resolve, and that is the ability to identify a range of audio to cut, copy, or delete using Timeline In and Out points, so that you can easily eliminate, move, or duplicate partial sections of audio without having to use the Razor Edit or Split Clip commands.

To delete a section of audio using In and Out points:

- 1 Set In and Out points in the Timeline to identify the range of audio you want to eliminate. If necessary, turn off the Auto Select controls of tracks to omit overlapping audio clips you don't want to delete from this operation.
- 2 Press the Backspace key to delete the section of audio and leave a gap, or press the Forward Delete key to delete the section of audio and ripple the Timeline to close the gap.



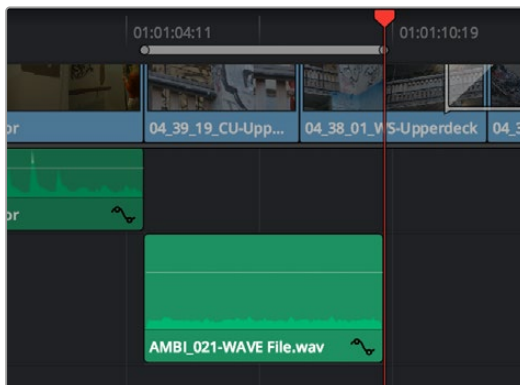
Setting In and Out points to identify a range of audio to delete



Deleting the audio using the Backspace key to leave a gap

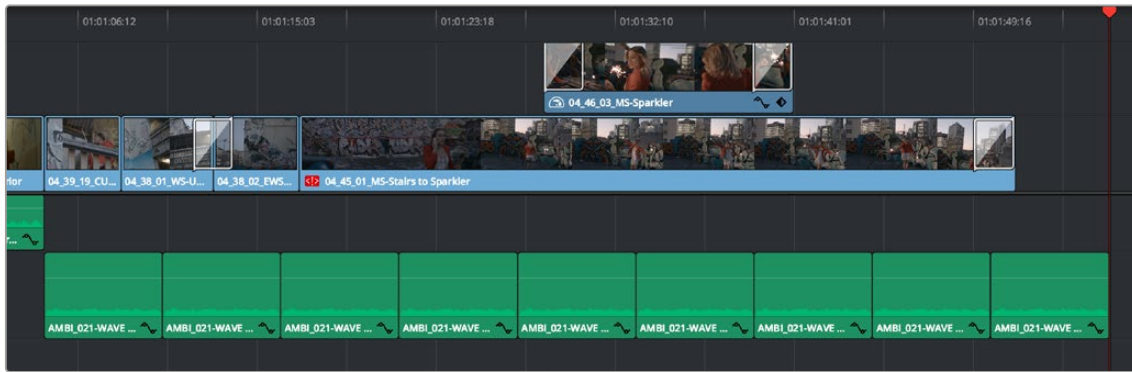
To copy a section of audio using In and Out points:

- 1 Set In and Out points in the Timeline to identify the range of audio you want to copy. If necessary, turn off the Auto Select controls of tracks with overlapping audio you don't want to copy; you can Option-click the Auto Select control of the audio track you're copying from to solo it, and you can Shift-click any video track's Auto Select control to turn them all off. In this example, we're copying some background ambience to continue building an ambience track.



Setting In and Out points to identify a range of audio to copy

- 2 Press Command-C to copy that section of audio.
- 3 Press Option-X to clear the Timeline In and Out points, and move the playhead to where you want to paste the copied section of audio.
- 4 Press Command-V to paste the copied audio. If you're looping a section of audio, you can paste many times to loop what you've copied.



Pasting the background ambience several times to loop it

Resizing Audio Clips in Subframe Increments

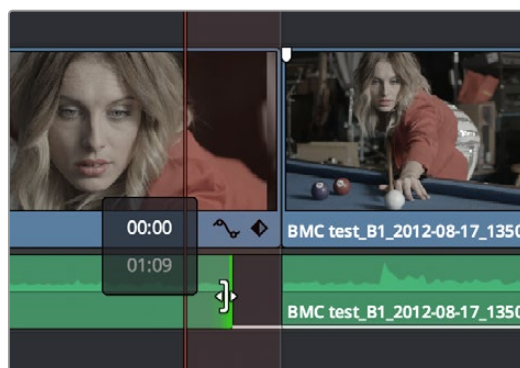
DaVinci Resolve lets you optionally make subframe audio adjustments to the In and Out points of audio clips in the Timeline.

Enabling and Disabling Subframe Editing

The “Align audio edits to frame boundaries” preference in the Editing panel of the DaVinci Resolve User Preferences lets you choose whether audio clip In and Out points align to whole frame boundaries, just like video clips. When this option is turned on, you cannot make subframe audio edits. When turned off (the default), you can.

Subframe Editing of In and Out Points

While you cannot move the playhead in subframe increments, you can resize audio clips in subframe increments by dragging an audio clip’s In or Out point in the Timeline, or by dragging an audio edit to perform a roll. This can be useful for trimming minute bits of audio such as pops, clicks, or vocalizations.



Resizing the Out point of an audio item in subframe increments, seen within a one-frame playhead shadow

Know that if you have Linked Selection turned on and you’re trying to resize a selected Video+Audio pair of items, the whole-frame resizing required for video prevents you from being able to resize the audio separately. This is easily solved by Option-Clicking to select the linked audio item by itself, at which point you can subframe resize it freely.

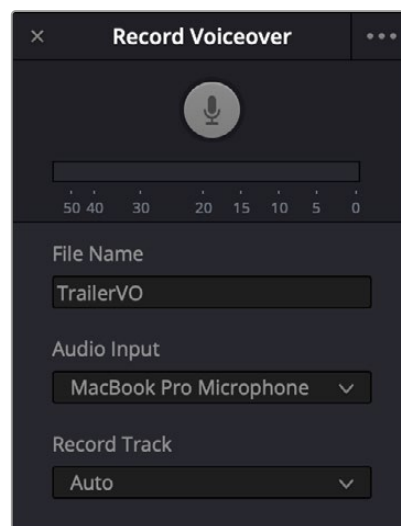
Also, if snapping is enabled, it may be impossible to make a subframe adjustment if you're too close to another edit point, a marker, or the playhead. In this case, pressing the N key to turn snapping off will solve the problem.

Eliminating Subframe Audio Edits

If you have a variety of subframe edits and need to eliminate these subframe adjustments, you can choose Timeline > Trim Audio Edits to Frame Boundaries. This conforms every edit on the timeline to frame boundaries on any clips that were edited to subframe precision. If the clip is longer than the nearest frame, it will be trimmed back, and if the start or end of a clip is close to a frame boundary and there is additional material available on the clip, it will be extended to reach the nearest frame boundary.

Record Voiceover Tool

In DaVinci Resolve 20, both the Cut and Edit pages have the ability to record voiceover directly to a track in the timeline using a simplified interface. For serious and precise ADR work, the ADR toolset in Fairlight will be more appropriate, but for simple VO and scratch tracks, the Record Voiceover tool is quick and convenient.



The Record Voiceover tool

To open the Record Voiceover tool, select Timeline > Record Voiceover, or select the Voiceover icon (the small microphone) on the Timeline Toolbar.

To Record Voiceover

- 1 File Name: Type in a descriptive file name for the new Voiceover file or just use the default name.
- 2 Audio Input: Choose which audio interface on your computer your microphone is connected to.
- 3 Record Track: Choose the audio track in the timeline you wish your voiceover track to be recorded to. Or select Auto to have DaVinci Resolve choose for you based on your current track layout.
- 4 Then place the playhead on the timeline where you want the audio to start.
- 5 Click the Record Icon to start recording directly to the track.
- 6 Click the Record Icon again to stop the recording.

When the recording stops, the playhead will automatically snap back to its original starting position for you to either play back the voiceover clip in place or to instantly record a new take by simply clicking on the Record Icon again. This new take will automatically overwrite the previous clip on the track. However, all previous takes will still be available in the Media Pool.

You can choose additional options in the 3-dot option menu of the Voiceover tool.

Input Monitoring: Turning this option on will play back the microphone audio as it's being recorded, albeit with some lag. This allows the voice over artist, or the audio engineer, the ability to hear the audio as it's being recorded. Selecting this option will make a Mic Monitor Level slider appear in the main Voiceover tool, which allows you to control how loud the playback will be in the headphones.

Enable 3 Seconds Countdown: Turning this option on will present a visual three-second countdown in the Timeline Viewer before the recording begins, allowing your talent a little time to get ready to speak. There is no audio beep for this countdown; it is a visual indicator only.

Mute Timeline Audio While Recording: Selecting this option will mute all audio tracks on the timeline while you are recording so that you do not hear any other sounds that could be distracting. If you want to mute just some of the tracks, you can do that manually using the mute buttons in the audio track timeline header.

Stereo Input: Selecting this option will record to a stereo track instead of mono, if you're using a stereo microphone.

Hide Options: Selecting this option will remove the File Name, Audio Input, and Record Track fields from the Voiceover tool, leaving only the Record icon to click on or off. This is useful if you don't want your voiceover talent to accidentally change your settings.

Audio Settings in the Inspector

Each clip or track has several audio-related parameters in the Audio panel of the Inspector.

For more details of these controls, see *Chapter 37, "Using the Inspector in the Edit Page."*

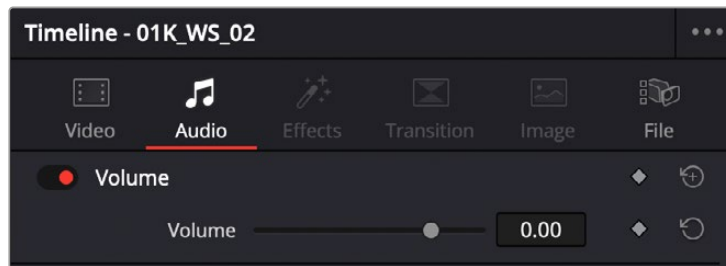
Setting Volume

Each audio clip, or audio item in the case of audio clips with linked audio on multiple tracks, has its own Volume level. This means that audio clips with multiple channels share a common Volume setting. There are several ways you can adjust these levels simply.

Adjusting Audio in the Inspector

Each clip has individual Volume parameters that are accessible in the Audio panel of the Inspector when one or more audio clips are selected.

Selecting an audio clip in the Timeline and adjusting its volume only alters that clip, which lets you set basic levels for individual clips in your program. The Volume control affects every channel within that clip simultaneously.

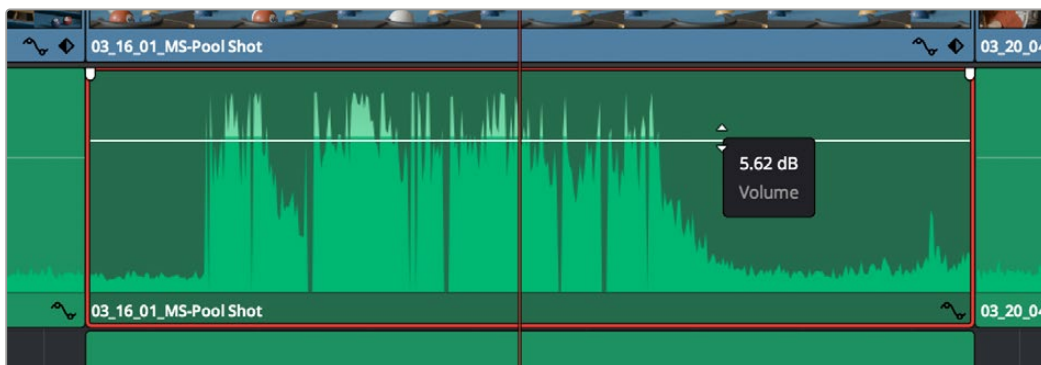


The Volume parameter available for audio clips in the Inspector

If you select multiple clips in the Timeline, then adjusting the Volume sliders or virtual sliders for all of them simultaneously will make a relative adjustment to all of the clips, preserving their offsets from one another. If you want to set all clips to the same level, then making a numeric adjustment will set all selected clips to the same absolute level.

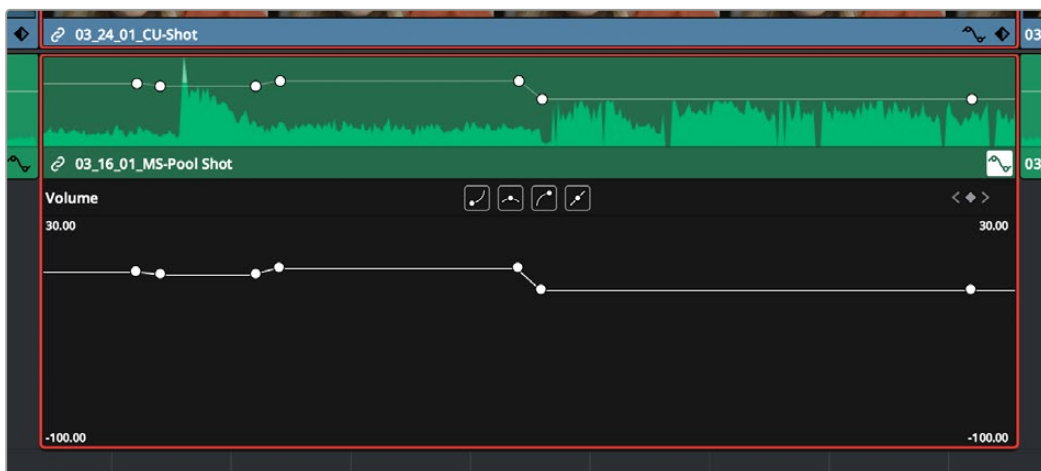
Adjusting Audio in the Timeline

Each clip (or item) of audio in the Timeline has a Volume overlay that lets you set that clip's level by simply dragging it up or down with the pointer. Holding the Shift key down while you drag allows finer adjustments. Option-clicking on the level creates a keyframe at that point to dynamically change levels. This overlay corresponds to the Volume parameter in the Inspector.



Dragging a Volume overlay to adjust the clip level

You can also click the Curve Editor button at the bottom right-hand corner of the audio clip, which opens the Audio Curve Editor. At the time of this writing, the only parameters you can edit in this Curve Editor is volume and pan.



Showing the Volume overlay in the Curve Editor

Adjusting Volume Using Keyboard Shortcuts

You can also adjust the volume of selected clips using keyboard shortcuts, even while the Timeline is playing. There are several ways you can set this up.

To adjust just one clip: Select that clip, and use one of the commands for changing volume.

To adjust any clip at the position of the playhead: Turn on Timeline > Selection Follows Playhead so that whichever clip intersects the playhead becomes selected, and use one of the commands for changing volume. If multiple clips are intersecting the playhead, the selected clip will be the one on the highest track.

To adjust multiple clips all together: Select all of the clips you want to adjust, all at once, and use one of the commands for changing volume. If the clips you select have differing Volume levels, these differences will be maintained as you make your adjustments.

The commands for changing volume are as follows:

To change volume in increments of 1dB: do one of the following:

- Clip > Audio > Increase Audio Level 1dB (Option-Command-Equals)
- Clip > Audio > Decrease Audio Level 1dB (Option-Command-Minus)

To change volume in increments of 3dB: do one of the following:

- Clip > Audio > Increase Audio Level 3dB (Option-Shift-Equals)
- Clip > Audio > Decrease Audio Level 3dB (Option-Shift-Minus)

Normalize Audio Volume Command

The Normalize Audio Levels command automatically adjusts the level of clips to a specific target level, and you can choose the method used to analyze each audio clip's levels to determine how to normalize each clip's volume. Options include a variety of loudness normalization algorithms specific to various international standards, which are useful for balancing the perceived overall loudness of several clips to one another, regardless of transient levels throughout each clip. You can also do Peak normalization, with options for both Sample Peak and True Peak.

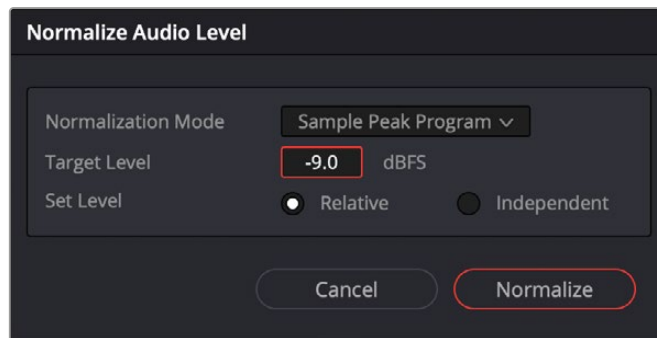
The various loudness options are designed to analyze an audio signal based on its perceived loudness to the listener, which results in a more accurate automatic balancing of different clips' audio levels to one another, regardless of transient peaks occurring throughout different clips.

The target peak meter now uses the BS.1774 standard for measuring maximum "true peak," which means that this meter is capable of measuring "inter-sample peaks," rather than only the peaks at each sample of a waveform. However, you still have the option to measure Sample Peak, which is the previous method of measuring the actual peak of the samples in a media file.

The change made by the Normalize Audio Volume command is only a volume adjustment; no dynamics are applied, so the result of using this command is that the loudest parts of each selected clip are going to match one another at the target level. This command is also available in the Fairlight page.

To normalize one or more selected audio clips:

- 1 Right-click one of the selected clips and choose Normalize Audio Levels. The Normalize Audio Level dialog appears.



The Normalize Dialog in the Edit page

- 2 Choose the Normalization Mode you want to use. You can choose among a variety of standardized loudness measurement algorithms, or Sample Peak, or True Peak.
- 3 Choose the reference level that you want to set the peak volume of the selected clips to match.
- 4 Choose how you want to set the level of multiple selected clips:

When Set Level is set to Relative, all selected clips are treated as if they're one clip, so that the highest peak level of all selected clips is used to define the adjustment, and the volume of all selected clips is adjusted by the same amount. This is good if you have a series of clips, such as a dialog recording, where the levels are consistent with one another, and you want to normalize all of them together.

When Set Level is set to Independent, the peak level of each clip is used to define the adjustment to that clip, so that the volume of every selected clip is adjusted by an amount specific to that clip. The end result may be a set of very different volume adjustments intended to make the peak levels of each audio clip match one another. This is good if, for example, you're trying to balance a series of different sound effects with one another that have very different starting levels.

For more information about loudness normalization, see *Chapter 182, "Audio Meters and Audio Monitoring."*

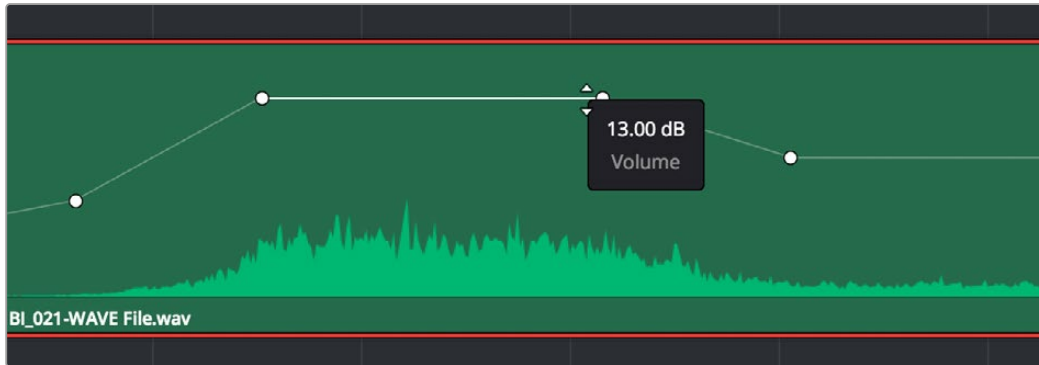
Keyframing Audio

There are two ways you can keyframe audio in the Edit page. You can use each audio clip's volume curve in the Timeline, or you can use the keyframe controls in the Inspector to animate the Volume parameter of individual clips as you would any other clip attribute, fading the level up or down, panning from left to right, or dynamically changing any one of a host of filter controls, in subframe increments, if necessary.

For more information on keyframing in the Inspector, see *Chapter 53, "Keyframing Effects in the Edit Page."* Any keyframes you create using the Keyframe controls of the Inspector automatically appear on the volume curve of that audio clip in the Timeline.

Overlay Controls for Volume

Each audio clip in the Timeline appears with a Volume overlay control on top of it, that by default starts out completely flat. Similar to such controls found in other applications, the level curve lets you alter each clip's levels, either overall, or dynamically using keyframes.



Volume curves adjusting the level of a clip's audio in the Timeline

Additionally, you can click any clip's Audio Curve Editor button, at the bottom right-hand corner of each audio clip, to open an audio-specific Curve Editor in which you can also keyframe volume.



The button for opening an audio clip's Curve Editor

How to Add and Adjust Volume Keyframes

Mixing audio by adding and adjusting individual keyframes can be a fast and effective way of balancing clip levels with one another, fixing level problems within a particular clip, or even creating simple mixes (although the mixing capabilities of the Fairlight page are considerably more robust). When manually editing any audio parameter curve, you can use the following procedures.

Methods of adding or selecting audio keyframes using the pointer:

- **To add keyframes to the Volume curve:** Hold the Option key down and click the curve to place a keyframe at that frame. You must add at least two keyframes to create an automated change in Volume.
- **To select one or more keyframes:** Click any keyframe to select it.
- **To select multiple discontinuous keyframes:** Command-click all keyframes you want to select, whether they're next to one another or not.
- **To select multiple contiguous keyframes:** Click the first keyframe you want to select, and then shift-click the last keyframe you want to select, and all keyframes between will also be selected.

Methods of adjusting keyframes in the Volume overlay (or curve) using the pointer:

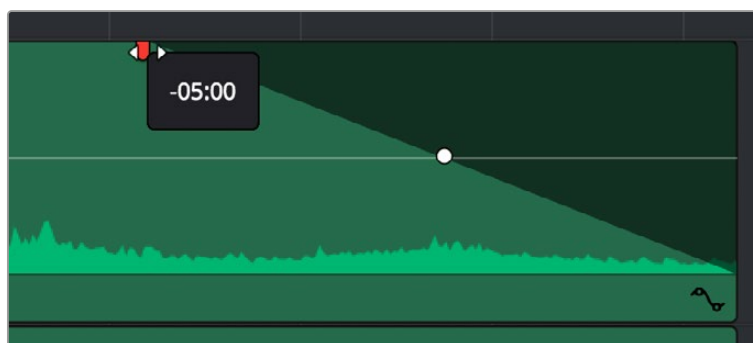
- **To adjust any curve segment:** Position the pointer over the overall segment for clips with no keyframes, or position it between any two keyframes, directly on top of the curve segment you want to raise or lower. When the Move cursor is displayed, click and drag up to raise the volume, or down to reduce the volume.
- **To adjust a keyframe in any direction:** Move the pointer over a keyframe so that the four-way cursor appears, and then click and drag up or down to change the volume, or side to side to change its timing. The timing of audio keyframes can be adjusted in subframe increments, for precision mixing.
- **To adjust a keyframe in only one direction:** Move the pointer over a keyframe so that the four-way cursor appears, press Shift, and click and drag in the intended direction of adjustment, either vertically to change the volume of the clip at that frame, or horizontally to move the keyframe to a different point in time. Once you start dragging a keyframe into a particular direction, keyframe movement is constrained in that direction until you release that keyframe. The timing of audio keyframes can be adjusted in subframe increments, for precision mixing.
- **To change one or more Linear keyframe to Ease In or Ease Out:** Eased keyframes create animated changes that begin slowly and accelerate to full speed, or slow down gradually to decelerate to a stop. This only works when you have two or more keyframes creating an animated effect. Select one or more keyframes, then right-click one of the selected keyframes and choose Ease In, Ease Out, or Ease In and Out, depending on which keyframe you're editing and the effect you want to create.
- **To change one or more eased keyframes to Linear:** Select one or more keyframes, then right-click one of the selected keyframes and choose Linear.

Methods of Cutting, Copying, Pasting, and Deleting keyframes:

- **To cut or copy, and paste one or more keyframes:** Make a selection of keyframes, and use the Cut (Command-X) or Copy (Command-C) key shortcuts. Then, move the playhead to where you want the first of the copied keyframes to start, and press Paste (Command-V).
- **To delete one or more control points from a curve:** Select the keyframe(s) you want to delete and press Backspace.

Audio Fade Handles

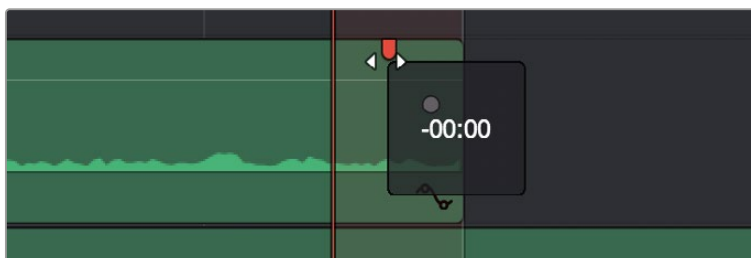
When you position the pointer directly over an audio clip, a pair of Audio Fade handles appear at the In and Out points. Dragging each of these handles towards the center of the clip lets you fade in the clip volume at the beginning of the clip, and fade out the clip volume at the end of the clip.



Audio Fade handles at either end of an audio clip

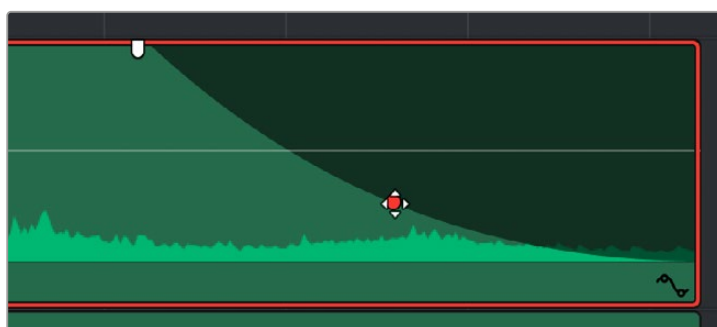
NOTE: When you import a Final Cut Pro X project, the fade handles for each clip automatically import as well.

Audio Fade Handles can also be adjusted in subframe increments, if necessary, to create a precise transition.



Adjusting an Audio Fade handle in subframe increments, seen within a one-frame playhead shadow

Once you've created a fade effect, you can adjust the curve of the fade by dragging the handle that appears right on top of the fader curve. Dragging the handle up and down affects the angle of the curve, and dragging the handle left and right affects the shape of the curve. In this way, you can create all manner of fade effects.



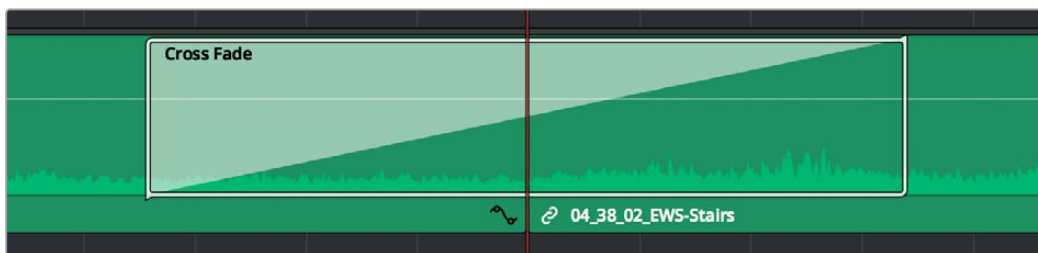
Adjusting the curve of the fade Fade effects can be created and edited on both the Edit and Fairlight pages.

Audio Crossfades

When you select an edit point with both video and audio components, and Linked Selection is enabled so that both the video and audio edit points are selected, then when you apply a video transition to an edit, a crossfade is added to the audio.

You can add Cross Fade transitions to any edit point between two audio clips that have enough handles similarly to how you add video transitions, by dragging and dropping from the Effects Library, by right-clicking an edit and choosing an option from the contextual menu, or by selecting an audio edit point and choosing Timeline > Add Audio Only Transition (Shift-T).

Cross Fade transitions are a quick and easy way to fade the volume of the outgoing clip down while simultaneously fading the volume of the incoming clip up, letting you create a smooth aural transition between two audio clips. If you need to do precision editing, the start and end points of a crossfade can be edited in sub-frame increments.



An audio Cross Fade transition applied between two clips

You can double-click a Cross Fade transition to open it into the Inspector, revealing the following parameters:

- **Duration:** The duration of the transition, shown in both seconds and frames.
- **Alignment:** A drop-down that lets you choose the transition's position relative to the edit point it's applied to. Your choices are "End on Edit," "Center on Edit," and "Begin on Edit."
- **Transition style:** You can choose -3dB , 0dB , or $+3\text{dB}$ to set both the Fade In and Fade Out levels to the same value. For more information on what these levels mean, see the following parameter.
- **Fade In/Fade Out levels:** There are three options that affect the incoming and outgoing halves of the Cross Fade effect independently. 0dB applies a linear fade (this is the default). $+3\text{dB}$ applies a boosted curve; when applied to both Fade In and Fade Out, this can compensate for diminished levels in the middle of a Cross Fade. -3dB applies an attenuating curve, which deliberately lowers the level of the Cross Fade.

Crossfades can be created and edited on both the Edit and Fairlight pages.

TIP: If you need an asymmetrical crossfade, that's accomplished by "checkerboarding" clips on two tracks with overlapping handles at the beginning and end, and using the Fader handles to create exactly the timing and curves necessary to create the effect you require.

Voice Isolation (Studio Version Only)

Voice Isolation uses an AI model trained for any type of human voice, male or female, young or old, which lets you completely remove loud, undesirable sounds from existing voice recordings with incredible results.

You can get rid of all kinds of background noise, from air conditioners or fans to jackhammers, restaurant background noise, music playing during a featured piece of dialogue, and more.

The Amount control adjusts the mix between the original source and the isolated voice. When set to 50, the mix is roughly equal. Values between 70 and 80 work well for natural results while strongly isolating the source.

Although Voice Isolation can be used on an audio track in real time, it cannot be used on live audio input. It can be used on any mono, stereo, or dual-mono audio track, but it isn't supported on tracks with more than two channels.



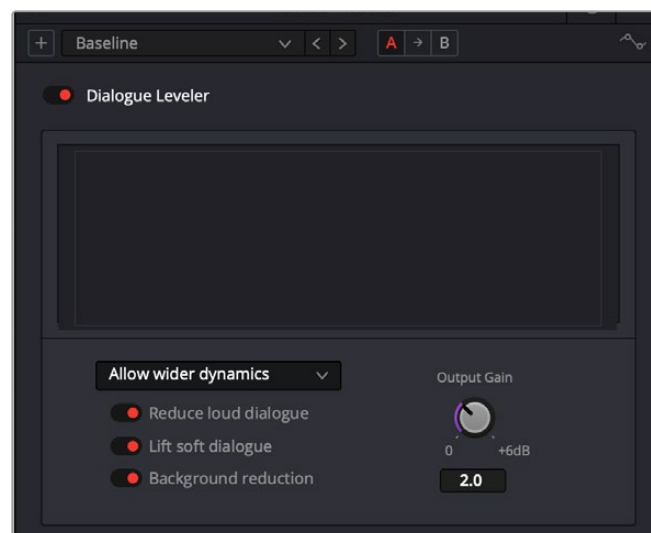
The Voice Isolation plugin

Dialogue Leveler

The Dialogue Leveler analyzes source material to detect dialogue and then “rides down” louder areas, “lifts up” softer areas, and lowers background sounds that are not dialogue.

It works without the typical “pumping” or other unwanted side effects of dynamics processors (compression/limiting) to produce results like those achieved through detailed, manual clip gain adjustments or “riding” the track with fader automation.

The Dialogue Leveler can often “fix and mix” a recording made with a single boom mic, where one character is closer than another or turns away from the mic and levels it out so that the original relative dynamics are preserved while increasing intelligibility and average level.



The Dialogue Leveler plugin

Although Dialogue Leveler can be used on an audio track in real time, it cannot be used on live audio input. It can be used on any mono, stereo, or dual-mono audio track, but it isn’t supported on tracks with more than two channels.

Dialogue Leveler controls:

Dialogue Leveler: Enables or disables the plugin.

- **Waveform Processing display:** This shows a scrolling waveform and a gray line indicating Dialogue Leveler processing. However, this display is not available when using the Dialogue Leveler found in the Inspector to process audio clips on the timeline.
- **Preset Menu:** This menu provides the following options:
 - Allow wider dynamics:** This is the default and is best for audio with a wider-ranging dynamic from loud to soft, and where the clip levels are medium to high.
 - More lift for low levels:** Select this option if the source has more low-level dialogue that you want to boost.
 - Lift soft whispery sources:** Select this option if the source has whispered dialogue and background noise.
 - Optimize moderate levels:** Select this option if the source is at medium levels throughout.
- **Reduce Loud Dialogue:** When enabled, louder dialogue is ridden downward when it peaks, acting like a “perfect limiter” where you don’t have to adjust the threshold or time parameters. Due to the “near real-time” functionality, analysis occurs prior to audible playback for optimal results.
- **Lift Soft Dialogue:** This setting finds and lifts low-level dialogue, evening out material that is quieter while more variable in level. Because the process is dialogue-focused, it doesn’t tend to raise background sounds unless they’re audible at the same time as the dialogue.
The Lift Soft Dialogue option is often the most useful of the three options, as it makes quieter lines of dialogue louder and naturally smooth without boosting background noise.
- **Background Reduction:** This option gently removes background noise based on the internal preset selected in the Preset menu.
- **Output Gain:** You can use this control to adjust the output level by clicking and dragging it or entering a value in the numeric field. The range of the Output Gain control of 0.0 to +6.0 dB.

TIP: Before using the Dialogue Leveler on a track, it can be useful to set an optimum baseline level for use with the default “Allow wider dynamics” preset by normalizing the audio.

For more information on normalizing audio, see *Chapter 176, “The Fairlight Inspector.”*

Music Remixer (Studio Version Only)

This DaVinci Neural Engine AI-based plugin splits music into individual stems: Vocals, Drums, Bass, Guitar, and “Other” (“everything else”), letting you creatively rebalance or remix the track.

You can use the level controls to adjust the volume of each stem and the Mute buttons to remove or bring them back into your mix. For example, you may want to remove or lower a vocal because it clashes with a featured piece of dialogue.



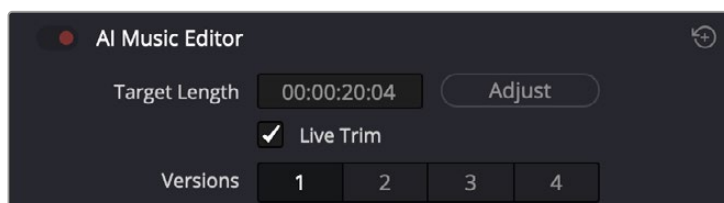
The Music Remixer plugin

Music Editor (Studio Version Only)

DaVinci Resolve 20 supports seamlessly extending or shortening music to the length of your edit. The new AI powered Music Editor can be accessed in the Audio tab of the Inspector. You can enter a desired duration and press the Adjust button in order to shorten or extend the clip to that duration.

NOTE: Music Editor is an “intelligent edit” that works effectively with beat-driven music (pop, dance, etc.). It is not intended for free form, ambient or non-beat driven material.

Music Editor does not use time compression/expansion or alter the pitch of the music, but instead uses editing techniques like a music editor might to fit the material to desired length to picture. It will look for logical places to create transitions based on whether you’re extending length (by repeating sections), or removing material, and attempt to use the ending or fade of the original composition if it times well. If the ending is abrupt, it’s up to you to manually create a fade out.

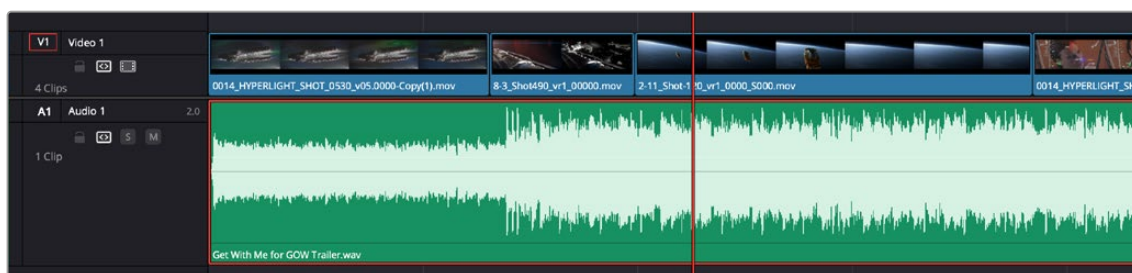


The Music Editor in the Audio Inspector

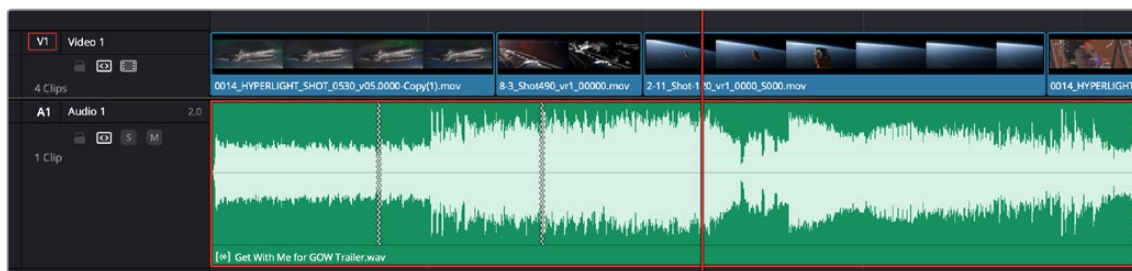
To Use Music Editor to retime a music clip:

- 1 Select the clip on the timeline you wish to retime. This clip does need to be beat-driven music.
- 2 In the Audio Inspector under AI Music Editor enter a new Target Length for the music duration you want.
- 3 Press the Adjust button.
- 4 Alternatively, you can check the Live Trim box to use the Trim tools to resize the clip and dynamically retime the music on the timeline. You can turn the checkbox off to return to normal editing (e.g., trimming) for that clip after the desired edit has been created.

Once the analysis is complete, four versions of potential edits you can use appear at the bottom, each with a different approach to the edit, numbered 1-4. Any subsequent editing can be done almost instantaneously. The edited pieces of music show indicators on the clip to mark the portions where cuts and transitions occur.



The initial music is too long



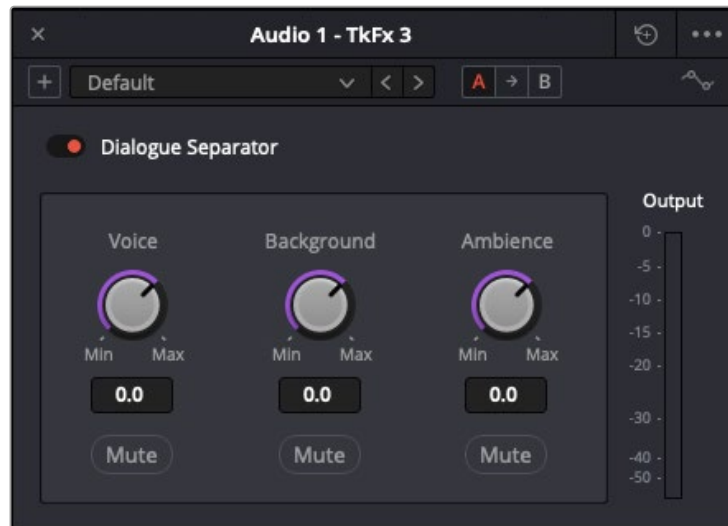
The Music Editor has retimed the audio clip to match the length of the video. Note the jagged lines on the audio clip indicating where the original was cut. In the waveform you can see the end of the music is retained in the retimed clip.

TIP: While timings can be quite close, sometimes they may not be exact. If you need to get an exact timing after working with AI Music Editor, you might use the Elastic Wave tool on the Fairlight page and choose the algorithm for general purpose (music).

The edited clip is actually a specialized Music Editor clip which allows you to further trim your edit. If you want to access the underlying edit, you can also right-click on the clip and decompose, exposing the edit points and any crossfades created. However, keep in mind that once decomposed, you can't trim the length of the clip as an entity as each editing snippet is now a separate clip.

Dialogue Separator Track FX (Studio Version Only)

Dialogue Separator uses DaVinci Neural Engine AI to give you individual control over the level of dialogue, background sounds, and the reverberant field or ambient room sound (“ambience”).



The Dialogue Separator plugin

The Dialogue Separator is useful when, for example, a source file has a great roomy sound, but there's music or crowd noise in the background. In this instance, you can rebalance the audio to reduce the background sounds or adjust the room sound to make the dialogue more intelligible.

Another use case would be when someone moves toward the camera in a roomy environment while speaking. You could use automation to rebalance the audio so that the ambient room sound gradually becomes “drier” as they get closer, without affecting the background.

Dialogue Separator is currently a mono-only plugin. If your source is stereo, you'll need to split it into dual-mono and process both mono tracks separately.

Ducker Track FX

This plugin lets you use the audible signal from a track or multiple tracks to lower the level of another.

Although it can be used in many creative ways, a common use of “ducking” is automatically lowering a music or sound effects bed so that dialogue or a VO track can be heard more prominently in a mix.

This is achieved without compressing the incoming signals. When set up correctly in the Ducker interface or via the Audio tab in the Inspector, your audio tracks can practically “mix themselves.”

In a simple use case, the Ducker uses level changes of a single incoming dialogue or VO track to decrease the volume of the sound effects bed of a noisy freeway.

In a more complex situation, such as a scene where three characters are conversing next to the same noisy freeway, a single Ducker plugin can use the level changes of each dialogue track to attenuate the traffic noise.

With only one input source (Dialogue 1), the Ducker would only be effective when the first character speaks. However, this new feature lets you add the other two dialogue tracks (Dialogue 2 and Dialogue 3) as input sources for attenuating the traffic noise.



The Ducker plugin

Using the Ducker

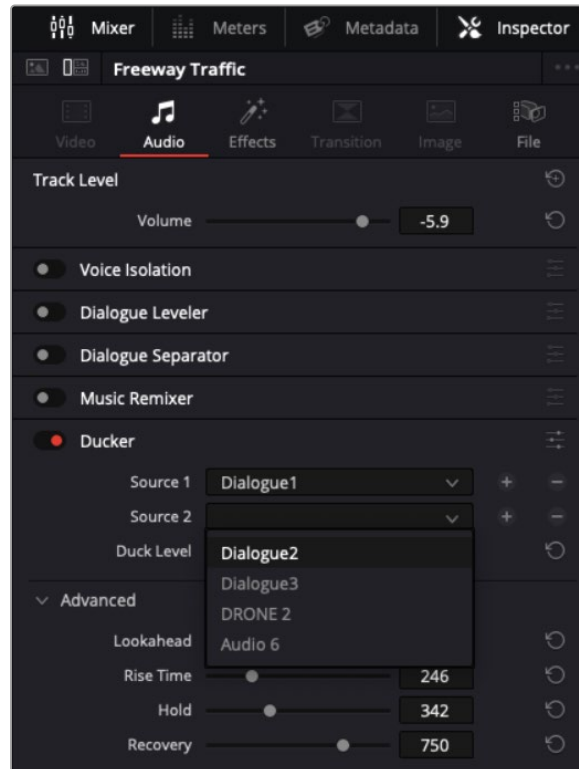
Continuing with the use cases mentioned above:

- 1 Choose the target track that you want to attenuate (the Freeway Traffic track), and open the Ducker.
- 2 Click the Source dropdown and select Dialogue 1 as the input source.
 - To use additional input sources, Command-click the other track names; in this example, Dialogue2 and Dialogue3.
 - Play a bit of your sequence, and adjust the duck level and other controls to taste.

Adding Input Sources via the Inspector

If you prefer working within the Inspector:

- Select the Freeway Traffic track, ensure the Ducker is switched on, and open the Inspector's Audio tab.
- Click the plus sign to the right of the Source 1 dropdown containing Dialogue1 to create a dropdown menu for Source 2, then choose Dialogue2 from the list.
- Click the plus sign to the right of the Source 2 dropdown to create a dropdown menu for Source 3, then choose Dialogue3 from the list.



Adding Ducker input sources in the Inspector

Ducker Controls

- **Duck Amount:** This control determines level attenuation on the target track in dB with a default value of 2.7 dB. Most of the time, a value between 2.0 dB and 5.0 dB works the best.
- **Lookahead:** Adjusts the amount of lookahead time in milliseconds, with a default value of 15ms. Lookahead allows the Ducker to start attenuating the destination track before the source track is audible.
- The larger the value, the more the Ducker will “pre-anticipate” the source track. If you want the Ducker to start lowering the sound before a line of dialogue or VO begins, you can experiment with higher values, but the default will usually produce good results.
- **Rise Time:** This determines how many milliseconds it takes for the Ducker to reach the amount of reduction specified by the Duck Amount. The default value is 10 milliseconds, which should work for most situations. Longer Rise Times may be useful with higher Duck Amount values and longer gaps in the source track.
- **Hold:** This parameter determines how long the target signal will be attenuated. The default is 150ms.
- **Recovery Time:** This value indicates how quickly the target signal will return to its previous level after attenuation. The default time value is 750ms, but setting the correct Recovery time is crucial to a natural-sounding ducking process, so you should try different values to find the best results.

Ducker Graph Display

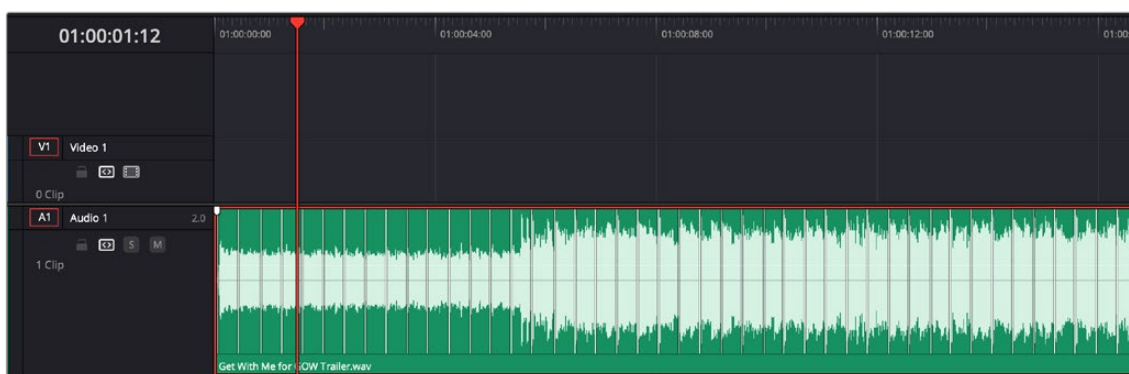
This area shows a gray line representing the incoming signal against the action of the Ducker, shown in yellow. Dips in the yellow line signify attenuation of the target track based on the Duck Amount in combination with the other parameters.

Beat Detector (Studio Version Only)

DaVinci Resolve can automatically detect beats for a piece of music on the timeline, and supports snapping to these beat marker locations. This can be especially useful when trying to make edits to the beat.

NOTE: The Beat Detector will only work effectively with beat-driven music, and with 4/4 or 3/4 time signatures at this time.

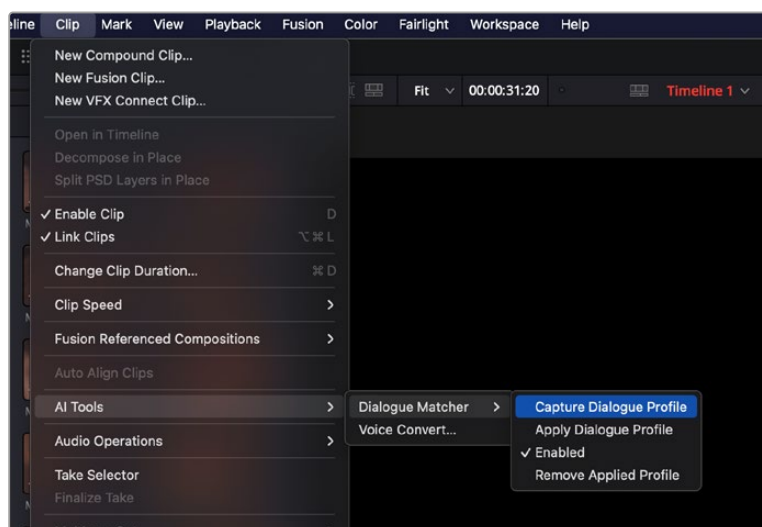
To detect beats, right-click on the timeline audio clip and select Detect Music Beats from the context menu. Once analyzed, the timeline clip displays beat indicator overlays that you can snap other timeline elements to.



A music track with the Beat Detector applied. The lined overlays match up with the beat of the music.

Dialogue Matcher (Studio Version Only)

Dialogue Matcher allows you to match one piece of dialogue to another, matching level, EQ, and reverb. This can help with situations where new recordings may not match the environment of existing ones. The Dialogue Matcher is a fully automated tool that only requires the user to capture a tone profile from a clip and then to apply it to another.



Capturing a Dialogue Profile from a clip using the main menu

To Use Dialogue Matcher:

- 1 Select a reference clip whose dialogue you want to match, right-click, and choose Clip > AI Tools > Dialogue Matcher > Capture Dialogue Profile.
- 2 Select the clip whose dialogue you want to change, right-click and choose Clip > AI Tools > Dialogue Matcher > Apply Dialogue Profile.
- 3 You can enable or disable the Dialogue Matcher on a clip by right-clicking and choosing Clip > AI Tools > Dialogue Matcher > Enable.
- 4 You can Remove the Dialogue Matcher on a clip by right-clicking and choosing Clip > AI Tools > Dialogue Matcher > Remove Applied Profile.

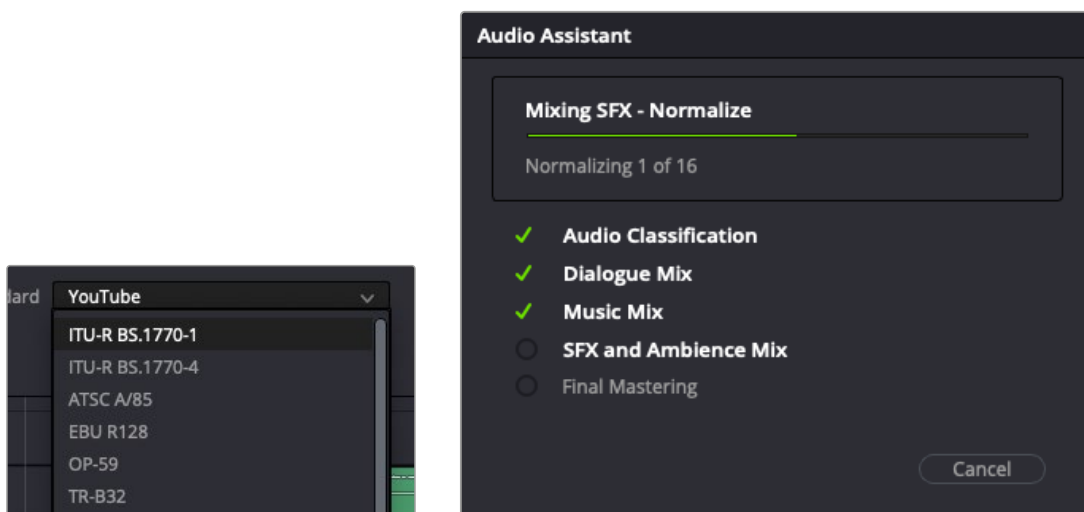
Audio Assistant (Studio Version Only)

With the Audio Assistant, DaVinci Resolve can automatically organize and color-code your tracks, even out your dialogue levels, and adjust the mixer faders to create a professional-quality mix of your music, sound effects, and dialogue.

During this process, your dialogue will also be cleaned up, and if needed, Voice Isolation and De-Essing will be applied. Ducking will be applied to music tracks, any required automation will be written, and the appropriate mastering and finalizing plugins will be added and adjusted to help achieve the correct loudness level for your chosen delivery standard.

To use AI Audio Assistant:

- 1 Open a timeline containing unmixed audio. The initial fader levels really don't matter as Audio Assistant will handle all fader adjustments during the mixing process.
- 2 To begin the automatic mixing process, choose Timeline > AI Tools > Audio Assistant from the Timeline menu, select the Delivery Standard for your mix from the dropdown menu, and click the Auto Mix button.
- 3 The Audio Assistant dialog changes to display the progress of the mixing process with an option to cancel it.



Audio Assistant dialog - Delivery Standard

Audio Assistant dialog - Mix Progress

- 4 When the mixing process has concluded, you can play back the mix and, if needed, make further adjustments or completely undo it.

NOTE: If a track is in the wrong category, right-click the corresponding track header and choose the correct category from the Track Category section of the contextual menu, and then rerun the Audio Assistant.

The Audio Mixer

The Audio Mixer on the Edit page is a simplified version of the Mixer on the Fairlight page, designed to provide a streamlined set of graphical controls you can use to set basic track levels (there is no track level fader automation on the Edit page), pan stereo audio at the track level, and mute and solo tracks as you work.

To open the Audio Mixer, do the following:

Click the Mixer button on the Interface toolbar.

The Audio Mixer exposes a set of channel strips with controls that correspond to the tracks in the Timeline, and each channel strip displays a number of audio meters equal to the number of channels within that track. By default, a Main 1 channel strip appears all the way to the right that lets you adjust the overall level of the mix. However, if you add subs and mains on the Fairlight page, those will appear at the right of the mixer as well.



The Audio Mixer, with four channel strips corresponding to the four tracks in the Timeline

NOTE: You cannot record automation in the Edit page. Comprehensive mixing controls with full automation recording are found in the Fairlight page.

Audio Mixer Controls

Each track's channel strip has the following controls:

- **Track Color:** Each track can be differently color-coded to help you keep organized. These colors also appear in the timeline track header and the Fairlight page.
- **Track Number:** The number of the timeline audio track corresponding to each channel strip appears here.
- **Track Effects:** Double-click on one of the track effects at the top of the fader controls to open.
 - FX:** Opens the Track Effects controls in the Inspector.
 - EQ:** Opens the Track Equalizer.
 - DY:** Opens the Track Dynamics.
- **Pan control:** Lets you pan a Mono track's audio from left to right, or invert a Stereo track's left and right audio channels, or do surround mixing.
- **Name:** The name of the audio track that channel strip corresponds to. If you've edited the audio track names in the Timeline, those names will appear here.
- **Solo:** Mutes all tracks other than ones that are soloed.
- **Mute:** Disables that audio track.
- **dB:** Shows you the volume, in decibels, that track is currently set to.
- **Fader:** Each track's vertical fader can be dragged with your mouse or other pointing device to adjust the volume of that track and perform automation recording. Dragging up increases volume, dragging down decreases volume.
- **Audio meters:** Audio meters to the right of each fader display the audio volume of all channels on that track during playback. Each channel strip has individual meters corresponding to the number of channels that track has been set to accommodate.

Mute and Solo Tracks For Output

When you use the Mute or Solo controls of the Audio Mixer, track audio is disabled both during playback and delivery for output. Make sure you have re-enabled any tracks you need before heading to the Deliver page. You can only modify mute and solo tracks on the Edit, Cut, and Fairlight pages.

Displaying Audio Meters

If you just want to see your program's levels, you can also switch to display the "Control Room" audio meters instead of the Mixer. How many audio meters appear depends on the current speaker configuration in the Video and Audio I/O panel of the System Preferences.

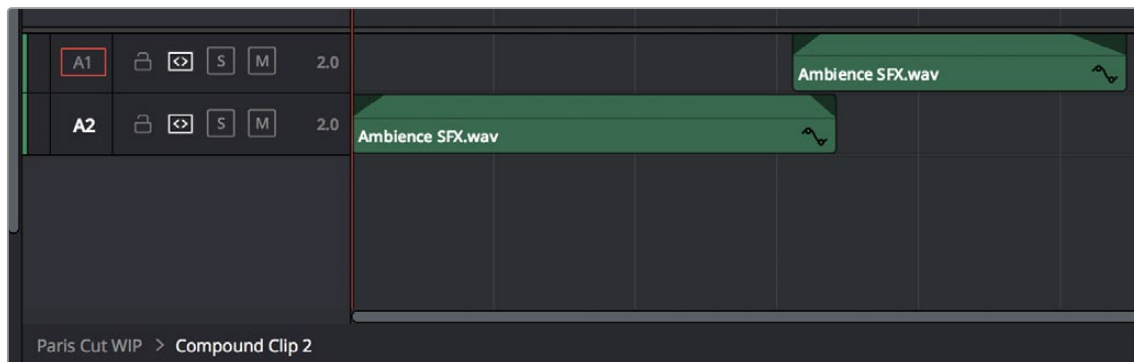
To show the Audio Meters:

Click the Mixer button on the Interface toolbar to display the audio panel and then choose Meters from the option menu at the upper right-hand corner.

Audio Compound Clips

DaVinci Resolve supports audio compound clips, which are created just like any other compound clip, by selecting multiple audio clips, right-clicking one of them, and choosing New Compound Clip. Alternately, compound clips with video clips may now contain multiple audio items as well.

When compound clips containing audio are opened in the Edit or Fairlight pages by right-clicking an audio compound clip and choosing Open in Timeline, breadcrumb controls appear beneath the Timeline that let you exit the compound clip and get back to the master Timeline.



Opening an audio compound clip

Audio Playback for Variable Speed Clips

Video/audio clips with variable speed effects applied to them can now play either pitch-corrected or un-pitch-corrected variable speed audio. An option in the Speed menu of the Retime controls lets you choose whether or not the audio is pitch-corrected.

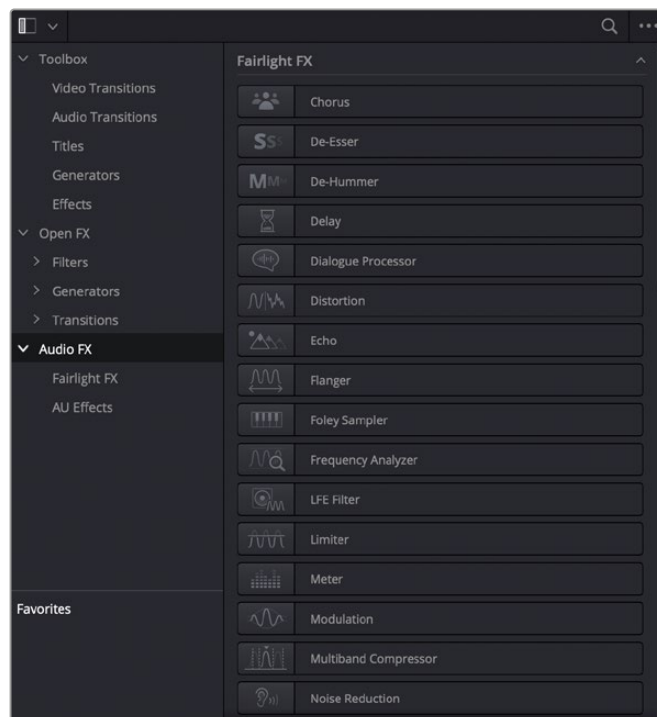
Using Audio Filters

DaVinci Resolve includes Fairlight FX, a set of DaVinci Resolve-specific audio plugins that run natively on macOS, Windows, and Linux, providing high-quality audio effects with professional features to all DaVinci Resolve users on all platforms. Additionally, DaVinci Resolve supports the use of third-party VST audio plugins on Mac OS X and Windows. On Mac OS X, DaVinci Resolve supports Audio Unit (AU) audio plugins. Once you install these effects on your workstation, they appear in the Audio FX panel of the Effects Library.

Audio plugins let you apply effects to individual audio clips or entire tracks worth of audio, to add creative qualities such as echo or reverb, or to take care of mastering issues using noise reduction, compression, or EQ.

Methods of applying audio filters to clips in the Edit page:

- **To apply an audio filter to a clip:** Drag any filter from the Audio FX panel of the Effects Library onto the clip in the Timeline you want to apply it to.
- **To apply an audio filter to multiple clips:** Select all of the clips you want to apply an audio filter to, then drag any filter from the Audio FX panel of the Effects Library onto any of the selected clips.

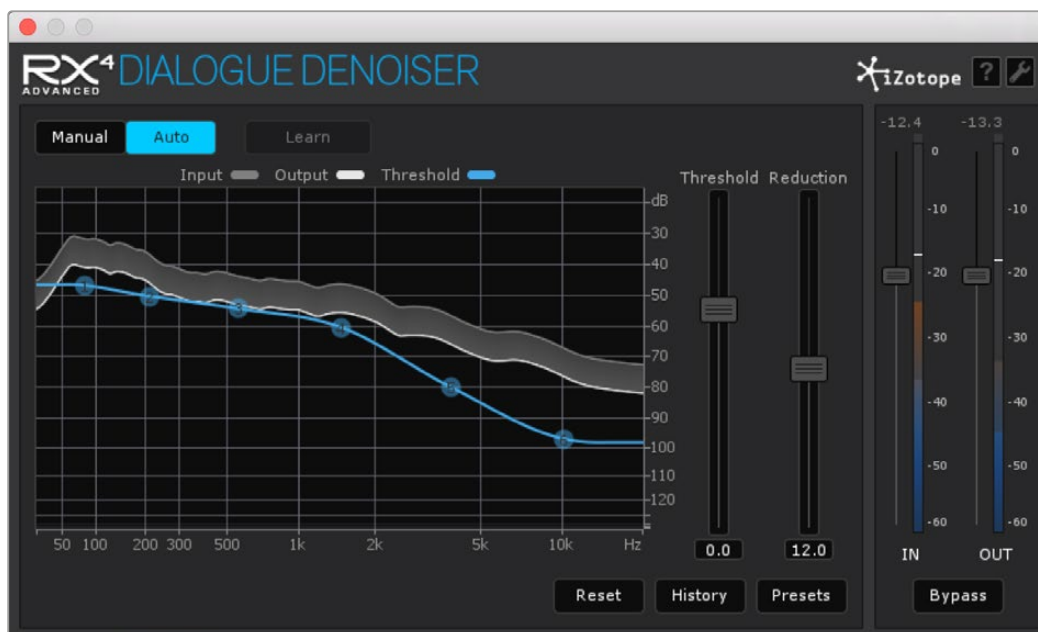


Audio filters in the Effects Library

To edit a clip's audio filters:

- Select that clip and open the Inspector. All audio filters applied to that clip appear under the Effects panel, in the Audio tab, with that filter's controls appearing directly in the Inspector.

Many VST and Audio Unit audio filters have a custom user interface that makes it much easier to manipulate that filter's controls. These can be opened from within DaVinci Resolve.



The custom audio filter interface for iZotope RX4

To expose a filter's custom controls:

- Click the Custom Control button (the button to the right of the trash can). The custom controls appear in a floating window. When you're finished adjusting the custom controls, close the window.



The button for opening a filter's custom control

Methods of working with audio filters in the Inspector:

- **To disable or re-enable a filter:** Click the toggle button at the far left of each filter's title bar.
- **To remove a filter:** Click the Trash Can button.

Once applied to a clip or track, audio filters can also be keyframed or automated just like volume and pan settings, to create dynamic audio effects that change over time.

Installing Audio Filters

VST effects aren't installed in a standard location, so it may sometimes be necessary to add a newly installed directory of VST plugins that you've just installed on your system. To help you deal with this, the Audio Plugins panel of the Preferences window has a list that lets you manually add and remove VST plugin directories, if necessary.

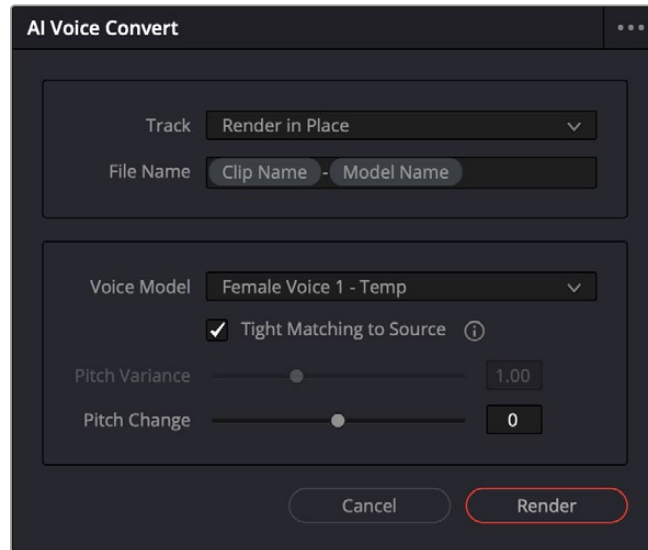
Once you've added one or more VST directories to the list, a second list underneath shows all audio plugins that are available within these directories. Each plugin on the list has a checkbox that shows whether or not it's currently enabled. Any VST plugins that cause DaVinci Resolve to crash while loading them during startup will be automatically disabled. You can use this list to see which plugins have been disabled, for troubleshooting purposes, and to reenabling such "blacklisted" plugins, by turning their checkboxes back on.

Voice Convert (Studio Version Only)

DaVinci Resolve has an incredibly sophisticated AI-driven voice conversion tool, that allows you to essentially swap one speaking voice for another. The use cases for this tool are wide reaching and even somewhat controversial, however this Voice Convert toolset can be an invaluable resource in a post production. It allows you to take a generated voice model that has been captured by DaVinci Resolve and "drive" that voice model using another voice recording as its foundation.

In simple cases, you can replace your own voiceover tracks with a voice that has more depth and gravitas to it, or even switch the gender of your own voice if you feel it's more appropriate for the subject matter. Perhaps you recorded audio in a noisy environment and wanted to make a pristine version instead by using your own voice as a model, to replace your original noisy tracks. In more complicated scenarios you can train your own voice convert model on your actor's voice patterns and use that to change and replace their previous voiceover, without having to get them back into the recording booth. Finally it is even possible to change the voice of an actor completely to another voice all while staying perfectly in sync with the picture. The inflection, pitch variations, and emotion of the original all remain; it just sounds like someone else.

Needless to say, using this powerful tool requires a certain level of responsibility on the part of the user. Privacy, copyright, and union laws and rules are all in flux at the moment around AI-generated content, and it's necessary to carefully navigate the ethics involved in its professional use.



The Voice Convert tool

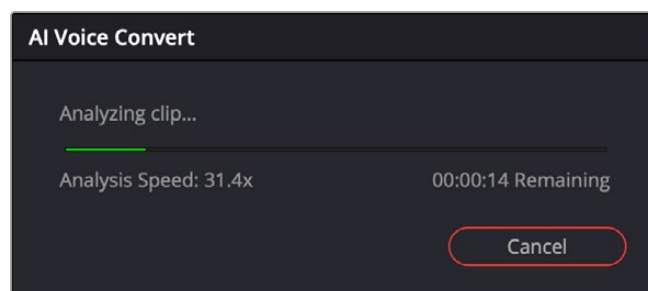
NOTE:~Using Voice Convert is very processor intensive and requires your system to have a minimum of 8 GB of video RAM before it will activate to avoid crashes and system instability. Even with a strong system creating a Voice Model will require prolonged intensive CPU and GPU power, so multi-tasking while it's working may be challenging.

Using Voice Convert

Like most of DaVinci Resolve's AI tools, using Voice Convert is very simple, with all of the complex processing involved handled by the voice model.

To use Voice Convert to change one voice to another:

- 1 Place the clip or clips you want to change the voice of in the timeline, and select them.
- 2 Right-click on the clip and select Voice Convert from the context menu, or select Clip > AI Tools > Voice Convert in the main menu.
- 3 Select the track that you want the new audio to appear in (including overwriting the original).
- 4 Select the file name you want the new audio to be named.
- 5 Select the Voice Model you want to use for the new audio (you can make your own custom voice models as explained below) and its parameters.
- 6 Click Render to start the Voice Convert process.



Voice Convert analyzing a clip

At this point the Voice Convert dialog will appear, showing you the progress of the conversion. Depending on the length and number of clips selected, this can take some time. When the process is done, the new audio will appear in the timeline in the exact same place as the original audio (or on the same place in another track if that option was selected). The new audio clips will also appear in the Media Pool.

Simply playback the timeline to audition the new voice. You can change the audio back to the original by right-clicking on a clip and selecting Revert to Original Voice from the context menu.

Tip: While you can drive voices with different accents, Voice Convert generally can't recreate another accent. For example if you have a voice model that is a speaker with a British accent and drive that voice with an American accent, you get more of an American version of the British accented speaker. But if you speak with a British accent when driving the British voice model, it will work well.

Voice Convert Settings

Each voice conversion requires selecting set up parameters to drive the model.

- **Track:** Select which track you want the new audio to be placed on. The default is Render in Place, which will overwrite the original audio.
- **File Name:** Set the file name for the new audio clip. You can enter a specific name, or use the metadata variables to create a template for file naming. By default the name is ClipName-(Voice Model Name).
- **Voice Model:** Select which voice model you wish to use for the new audio. There are some built-in options that come with DaVinci Resolve, and any custom options you create will also be available here.
- **Tight Matching to Source:** Check this box to tightly follow the pitch, intonation and volume variance of the source voice. Uncheck this box to have a looser interpretation using the Pitch Variance tool.
- **Pitch Variance:** When the Tight Matching to Source box is unchecked, this tool becomes active. Lower values create a more monotone interpretation, while higher values have more variance in pitch.
- **Pitch Change:** Use this slider to create a deeper (negative values), or higher (positive values) voice based on the model. The values are in semitones and can be adjusted in 100 cents (.1 semitone) increments.

Creating Your Own Voice Models

It is possible to create your own voice models for use with Voice Convert. These can be of your own voice or even someone else's that you have the rights to. This section goes over some best practices to get a clean and usable voice model.

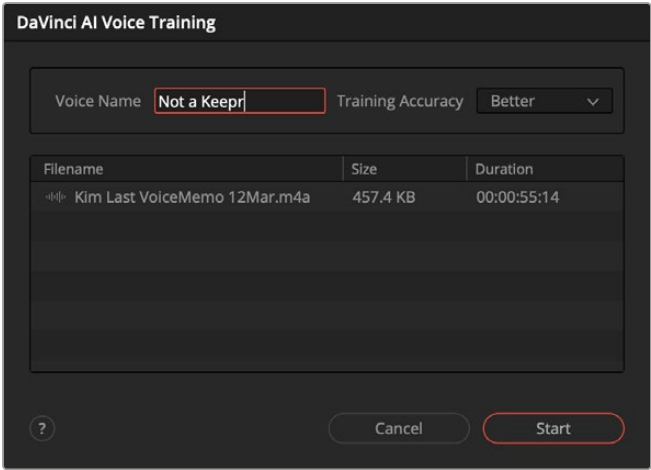
Preparing the Voice Model Assets

You will get the best results using ten minutes or so of clean, high-quality recordings of the voice to be modeled. The audio should be free of excessive ambiance and noise, with little to no dynamics processing. You will want consistent material that sounds natural. It can be useful to have a variety of emotions expressed as well, if you want to hear those results for the driven voice. Treat it like a voice over audition; show your range.

These voice assets do not necessarily need to be in one long clip, though if there are a lot of separate files, it can be helpful to consolidate them all in a single bin for organization. The multiple clips should be of consistent quality, without audio variations in the recorded sources.

Creating the Voice Model

To generate your own voice model, select one or more clips for analysis in the Media Pool, right-click and choose AI Tools> DaVinci AI Tools Voice Training from the context menu. If this is the first time you've run this tool, DaVinci Resolve will open up the Resolve Extras Download Manager to install the required support files.

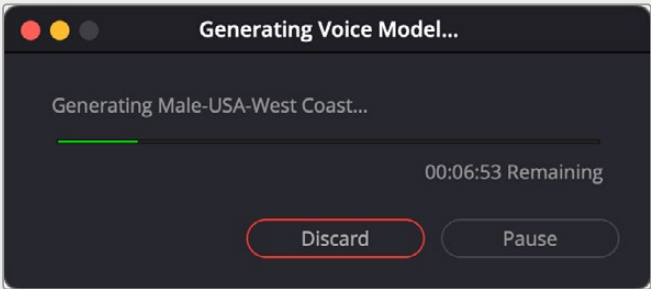


Voice Model Generation

Initially a warning splash screen appears, to ensure you assume full responsibility for how you will use this tool. Once you hit accept, the DaVinci AI Voice Training dialog appears, and there are two options to set.

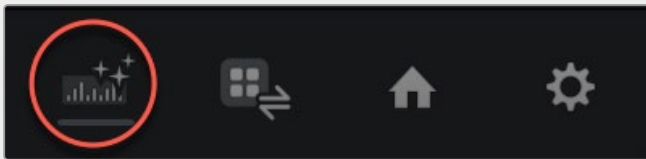
- **Voice Model Name:** Enter a descriptive name for your new model.
- **Quality:** Choose Faster or Better mode for analysis. Better takes about 3x the time as faster, and 10 minutes or so of material can take a few hours to complete depending on your computer's power.

Initially creating a voice model launches a foreground process, then once the initial analysis is complete, it will switch to a background process, allowing you to continue to use DaVinci Resolve for other purposes or other apps while it finishes.



Voice Model progress

You can see the progress of the model in the Voice Model Generation Progress indicator in the bottom-right corner, or by selecting Workspace > Voice Model Progress.



The Voice Model Generation Progress indicator will slowly fill blue as the voice model finishes. Hover over the top to get an ETA and progress finished tooltip.

You can also alternatively choose to pause or discard the in-progress model instead. Quitting DaVinci Resolve will also pause the background process, and it will automatically start up again the next time DaVinci Resolve is launched.

Once the model is complete, the new voice preset will appear for use in the Voice Convert Dialog.

The Fairlight Page

The audio controls of the Edit page are geared more towards simple mixing to have sensible levels as you work putting a program together. For comprehensive audio sweetening, mixing, automation, and mastering controls, the Fairlight page is only one click away.

For more information, see *Part 13, "Fairlight."*